MAHATMA GANDHI UNIVERSITY

UNDER GRADUATE VOCATIONAL PROGRAMMES (HONOURS)

SYLLABUS

MGU-B.VOC. (Honours)

(2025 Admission Onwards)

Faculty: Technology and Applied Sciences

Expert Committee: Emerging Technology and Digital Proficiency

Programme: B.Voc.(Honours) Software Development and System Administration

Mahatma Gandhi University

Priyadarshini Hills
Kottayam – 686560, Kerala, India

Preface

The Outcome-Based Syllabus (OBS) presented here is designed for the **Bachelor of Vocation (B.Voc.)(Honours) Programme in Software Development and System Administration**. This curriculum is the outcome of in-depth research, thoughtful analysis, and collaborative input from academic professionals, industry experts, and other key stakeholders. It is structured to align with the evolving demands of the information technology sector and the broader landscape of higher education.

The primary objective of this syllabus is to equip students with the essential knowledge, practical skills, and professional competencies required to succeed in the domains of software development and system administration. By embracing an outcome-based approach, the syllabus fosters critical thinking, technical proficiency, adaptability, and innovative problem-solving—qualities vital for meeting the challenges and seizing the opportunities of the digital age.

- **Industry Driven Curriculum:** The syllabus is tailored to reflect current trends, technologies, and best practices in the IT industry, ensuring that graduates are well-prepared for roles in software engineering, system operations, IT support, DevOps, cloud services, and more.

- **Emphasis on Core Concepts:** Foundational subjects such as programming, databases, operating systems, and networking form the core of the curriculum, providing students with a robust theoretical base for future advancement.

- **Experiential and Practical Learning:** Hands-on learning through lab work, live projects, simulations, and internships is a central component, allowing students to apply concepts in practical settings and develop job-ready skills.

- **Cross Disciplinary Integration:** The syllabus integrates elements from mathematics, business processes, communication, and emerging technologies, enabling a comprehensive understanding of IT systems and their real-world applications.

- **Continuous Evaluation and Feedback:** Ongoing assessment, project reviews, and feedback mechanisms support academic progress and encourage reflective learning, helping educators adapt strategies to enhance student outcomes.

- **Flexibility and Adaptability:** The syllabus is designed to remain flexible, allowing for updates in response to technological innovation, evolving job roles, and stakeholder feedback—thus ensuring long-term relevance.

We believe that this Outcome-Based Syllabus will serve as a valuable foundation for shaping competent, ethical, and future-ready professionals in the fields of software development and system administration. It is our hope that this program fosters a lifelong commitment to learning, growth, and excellence in the technology-driven world.

Expert Committee , External Experts and Syllabus Workshop Participants

Expert Committee – Emerging Technology and Digital Proficiency			
Sl:No	**Name of Participant**	**Designation**	**Address**
1	Dr. Leena C.Sekhar(Convener)	Associate Professor of Computer Applications & HOD	MES College, Nedumkandam
2	Dr. Shereena V. B.	Associate Professor of Computer Science	MES College, Nedumkandam
3	Dr. Antu Annam Thomas	Assistant Professor of Computer Applications & HOD	Mar Thoma College, Thiruvalla
4	Dr. Jina Varghese	Assistant Professor of Computer Applications	Baselios Paulose II Catholicose College,Piravom
5	Meera Nair	Assistant Professor of Computer Applications	NSS College, Rajakumari
6	Soumya M V	Assistant Professor of Computer Science	SAS SNDP Yogam College, Konni
7	Resmi Cherian	B.Voc Nodal Officer	CMS College, Kottayam
8	Christina Merin Babu	Assistant Professor (B.Voc)	Mar Athanasius College, Kothamangalam
9	Amruthavarshini R	Assistant Professor (B.Voc)	St. Xaviers College for Women, Aluva
10	Raji S. Pillai	B.Voc Nodal Officer	St. Theresa's College, Ernakulam
11	Spasiba Raveendran	Assistant Professor of Computer Science	SAS SNDP Yogam College, Konni
12	Syam S.	Assistant Professor of Computer Applications	Sree Narayana Arts & Science College, Kumarakom,Kottayam

13	Dr. Abahayadev M.	Assistant Professor of Computer Science	MES College, Nedumkandam
14	Shanavas C.T.	Assistant Professor of Computer Science	MES College, Marampilly
External Experts			
15	Mr. Aswin Santhakumar	Managing Director Managing Partner	SWATHANTRA INNOVATION PVT.LTD CELLAR INNOVATIVE DEVELOPERS Email : aswinsan127@gmail.com Contact : +91 8157983556
Syllabus Workshop Participants			
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Syllabus Index

Name of the Major Subject: Software Development and System Administration

Semester: 1

Course Code	Title of the Course	Type of the Course	Credit	Hours/ Week	Hour Distribution /week		
					L	P	O
MG1SDCSDS100	Web Designing & Internet Programming	SDC	4	5	3	2	0
MG1SDCSDS101	Programming in C	SDC	4	5	3	2	0
MG1SDCSDS102	Mathematics for System & Software	SDC	4	4	4	0	0
MG1MDCSDS100	Digital Marketing & Techniques using AI	MDC	3	3	3	0	0
MG1SDCSDS103	On-the-Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Semester: 2

Course Code	Title of the Course	Type of the Course	Credit	Hours/ Week	Hour Distribution /week		
					L	P	O
MG2SDCSDS100	Computer System Design & Architecture	SDC	4	4	4	0	0
MG2SDCSDS101	Database Management Using MySQL	SDC	4	5	3	2	0
MG2SDCSDS102	PHP Programming	SDC	4	5	3	2	0
MG2MDCSDS100	Web Designing	MDC	3	4	2	2	0
MG2SDCSDS103	On-the-Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Semester: 3

Course Code	Title of the Course	Type of the Course	Credit	Hours/ Week	Hour Distribution /week		
					L	P	O
MG3SDCSDS200	Software Engineering & Software Testing Principles	SDC	4	4	4	0	0
MG3SDCSDS201	Data structure using C++	SDC	4	5	3	2	0
MG3SDCSDS202	Desktop Application Development using .NET & SQL Server	SDC	4	5	3	2	0
MG3MPCSDS200	Database Management System (DBMS)	MPC	4	5	3	2	0
MG3MDCSDS200	E-governance and Digital Kerala System	MDC	3	3	3	0	0
MG3SDCSDS203	On-the-Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Semester: 4

Course Code	Title of the Course	Type of the Course	Credit	Hours / Week	Hour Distribution /week		
					L	P	O
MG4SDCSDS200	Artificial Intelligence and Machine Learning	SDC	4	4	4	0	0
MG4SDCSDS201	Python Programming	SDC	4	5	3	2	0
MG4SDCSDS202	Operating System & Linux Administration	SDC	4	5	3	2	0
MG4MPCSDS200	.NET & SQL Server – Desktop Application Development	MPC	4	5	3	2	0
MG4SECSDS200	MIS Deployment in IT	SEC	3	3	3	0	0
MG4VACSDS200	Social Impact of Technologies	VAC	3	3	3	0	0
MG4INTSDS200	Internship	INT	2	60 Hours			

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Semester: 5

Course Code	Title of the Course	Type of the Course	Credit	Hours/ Week	Hour Distribution /week		
					L	P	O
MG5SDCSDS300	Cloud Computing & DevOps	SDC	4	4	4	0	0
MG5SDCSDS301	React.js	SDC	4	5	3	2	0
MG5SDESDS300	Node.js	SDE (Elective Papers – Select one)	4	5	3	2	0
MG5SDESDS301	ASP .NET Model View Controller Application						
MG5SDESDS302	Enterprise Database Administration with Advanced SQL Server						
MG5SECSDS300	Computer Hardware Maintenance & Troubleshooting	SEC	3	4	2	2	0
MG5MPCSDS300	Programming with Python	MPC	4	4	4	0	0
MG5VACSDS300	Cyber Laws & Online Security	VAC	3	3	3	0	0

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Semester: 6

Course Code	Title of the Course	Type of the Course	Credit	Hours/ Week	Hour Distribution /week		
					L	P	O
MG6SDCSDS300	Mobile Application Development	SDC	4	5	3	2	0
MG6SDESDS300	Java Programming	SDE(Elective Papers -Select one)	4	5	3	2	0
MG6SDESDS301	Data Analytics Using Python						
MG6SECSDS300	Big Data Analytics	SEC	3	3	3	0	0
MG6MPCSDS300	Introduction to Machine Learning	MPC	4	4	4	0	0
MG6VACSDS300	Methodologies for Computational Research	VAC	3	3	3	0	0
MG6PRJSDS300	Project	PRJ	4	8	0	8	0

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Semesters: 7 & 8

B.Voc. Honours							
Course Code	Title of the Course	Type of the course	Credit	Number of Days	Credit Distribution		
					L	P	O
MG7APPSDS400	Apprenticeship	APP	28	280	0	28	0
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

B.Voc. Honours with Research							
Course Code	**Title of the course**	**Type of the course**	**Credit**	**Number of Days**	**Credit Distribution**		
					L	**P**	**O**
MG7RINSDS400	Research Internship	RIN	20	200	0	20	0
NA	Online	SDC	4	NA			
NA	Online	SDC	4	NA			
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Job Roles and Qualification Packs for Certificate, Diploma, Bachelor's, and Honours Degrees.

Job Roles	NHEQF Level	QPs Aligned	Sector Skill	Year
Web Developer	Level 4.5	1. SSC/N0501, v3.0 2. SSC/N0503, v3.0 3. DGT/VSQ/N 0101, v1.0	IT/ITeS	First Year
Junior Software Developer	Level 5.0	1. SSC/Q0508 2. SSC/N0506, v3.0 3. DGT/VSQ/N 0101, v1.0	IT/ITeS	Second Year
Software Developer	Level 5.5	1. SSC/Q0501	IT/ITeS	Third Year
IT Application Engineer	Level 6.0	1. NIE/ITS/N14082	IT/ITeS	Fourth Year

SEMESTER I

Semester: 1

Course Code	Title of the Course	Type of the Course	Credit	Hours/ Week	Hour Distribution /week		
					L	P	O
MG1SDCSDS100	Web Designing & Internet Programming	SDC	4	5	3	2	0
MG1SDCSDS101	Programming in C	SDC	4	5	3	2	0
MG1SDCSDS102	Mathematics for System & Software	SDC	4	4	4	0	0
MG1MDCSDS100	Digital Marketing & Techniques using AI	MDC	3	3	3	0	0
MG1SDCSDS103	On-the-Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**Web Designing & Internet Programming**
Type of Course	**SDC**
Course Code	**MG1SDCSDS100**
Course Level	**100**
Course Summary	This course introduces students to the core technologies used in modern web development and user experience design. It covers HTML5 and CSS3 for content structuring and styling, JavaScript and jQuery for client-side interactivity and UI/UX concepts with hands-on prototyping using tools like Figma. Through this course, students will gain essential skills to build responsive, interactive, and user-friendly website.

Semester	1	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand HTML5 and CSS3 to design and develop well-structured, semantic, and responsive web pages.	U	2,10

2	Demonstrate proficiency in creating dynamic and interactive user experiences using JavaScript and jQuery.	A	1,2,10
3	Apply CSS3 layout techniques and UI/UX principles to enhance design and usability.	A	2,3,4,10
4	Design and implement a working website integrating front-end skills, UI/UX tools	A	1,2,3,4,5,9,10
Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**0**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 2	**3**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
CO 3	**0**	**1**	**2**	**3**	**0**	**0**	**0**	**0**	**0**	**1**
CO 4	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**0**	**3**	**1**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I - HTML5 and CSS3 Basics (15 Hours)**			
	1.1	Introduction to HTML5:Document structure, Tags and Elements	2	1
	1.2	Text Formatting and Lists (ordered, unordered, description) Adding Multimedia: <img>, <audio>, <video>, <iframe>	2	1

	1.3	Tables and Forms: <form>, input elements, <textarea>, <select>, <fieldset>, validation attributes	3	1
	1.4	Semantic Elements: <header>, <footer>, <article>, <section>, <nav>	3	1
	1.5	Introduction to CSS: Types (inline, internal, external), Selectors (ID, class, element)	2	1
	1.6	Box Model: Margin, padding, border, content Responsive Web Design Basics	3	1
2	**Module II - CSS3 and UI/UX Concepts (15 Hours)**			
	2.1	Positioning: static, relative, absolute, fixed, z-index Display and Float Properties	3	3
	2.2	CSS3 Effects: Transitions, Animations, Transformations (rotate, scale)	3	3
	2.3	Layout Techniques: Flexbox, Grid System	2	3
	2.4	Introduction to UI/UX Design : Difference between UI and UX, User-Centric Design Principles	3	3
	2.5	Figma interface, tools, and prototyping basics	4	3
3	**Module III - JavaScript and jQuery (15 Hours)**			
	3.1	Introduction to JavaScript, Basic Data Types & Variables, Control Structures	3	2
	3.2	Functions, Arrays and Objects, Event-Driven Programming	2	2
	3.3	Introduction to jQuery, jQuery Basics, Working with Events	3	2
	3.4	Querying the Document, Modifying the Document	3	2

	3.5	networking with jQuery, Working with Forms	4	2
4	**4.1**	**(Practical Content)** 1. Write a program to illustrate heading tags 2. Write a program to illustrate all text formatting tags 3. Create your class timetable using table tag. 4. Write a program to illustrate nested lists and description list 5. Write a program to illustrate the iframes in HTML. 6. Write a program to illustrate a different types of selectors in CSS using external CSS 7. Use all the CSS (inline, internal and external) to format college web page that you have created. 8. Write a program to demonstrate CSS box model. Use inline and internal CSS 9. Write a program to implement the semantic and structural elements in html. 10. Write a program to create an HTML registration form and validation using JavaScript. 11. Write a program to illustrate the events in JavaScript.	30	4

		12. Use jQuery to create form with input validation and dynamic content display 13. Write a program to demonstrate fade in and fade out of any HTML element using. Jquery 14. Design a Webpage for your college containing description of courses, department, faculties, library etc. using HTML/CSS/JS/Jquery and design.		
5	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment**
	A. Continuous Comprehensive Assessment (CCA) **CCA for Practical (30 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva

	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination(70 Marks,3 hrs)** Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES

1. Freeman, E., & Robson, E. (2014). Head First HTML and CSS (2nd ed.). O'Reilly Media.
2. Flanagan, D. (2020). JavaScript: The definitive guide (7th ed.). O'Reilly Media.
3. Beaird, J., & George, J. (2020). The principles of beautiful web design (4th ed.). SitePoint.

SUGGESTED READINGS

1. Jon Duckett,-Web Design with HTML, CSS, JavaScript and jQuery Set 1st Edition, Wiley; 1 edition (July 8, 2014).
2. Laura Lemay, Rafe Colburn, Jennifer Kyrnin - Mastering HTML, CSS &Javascript Web Publishing,First edition, BPB Publications;.
3. Hartson, R., & Pyla, P. S. (2019). The UX book: Agile UX design for a quality user experience (2nd ed.). Morgan Kaufmann.

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Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**				
Programme	**B.Voc.(Honours)Software Development and System Administration**				
Course Name	**Programming in C**				
Type of Course	**SDC**				
Course Code	**MG1SDCSDS101**				
Course Level	**100**				
Course Summary	This course is designed to build problem-solving abilities using the C programming language. It highlights the hands-on use of key C features such as control statements, loops, arrays, and functions. By the end, students will be capable of creating efficient solutions using C programming.				
Semester	1	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Recall and understand fundamental programming principles.	K	1,2
2	Understand C Programming Basics such as Datatypes and variables, Different types of operators.	U	1,2,3

3	Devise C programs using the concept of Decision statements and loop control statements.	A	1,2,3,10
4	Implement modular programs using loops, functions, recursion, structures, unions, and dynamic memory allocation.	A	1,2,3,9,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
CO 2	**3**	**2**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
CO 3	**2**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
CO 4	**2**	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**2**	**3**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I – Fundamentals of Programming (15 Hours)**			
	1.1	Introduction to Programming: Computer program. Classification of computer languages: machine, assembly and high-level languages.	4	1
	1.2	Understanding basic Problem- Solving Tools: Algorithms: Definition & its attributes, examples.	4	1
	1.3	Flowchart: Definition & its attributes, symbols, examples.	4	1
	1.4	Types of errors- Syntax errors, Logical errors and Runtime errors.	3	1

2	**Module II - C Programming Syntax and Operators (15 Hours)**			
	2.1	C Character Set, Delimiters, Types of Tokens, C Keywords, Identifiers, Constants, Variables, Rules for defining variables	4	2
	2.2	Data types, C data types, Declaring and initialization of variables, Type modifiers, Type conversion, Operators and Expressions-	4	2
	2.3	Properties of operators, Priority of operators, Comma and conditional operator, Arithmetic operators, Relational operators,	3	2
	2.4	Assignment operators and expressions, Logical Operators, Bitwise operators.	2	2
	2.5	Input and Output in C – Formatted functions, unformatted functions, commonly used library functions.	2	2
3	**Module III - C Programming Control Structures and Data Handling (15 Hours)**			
	3.1	Decision Statements If, if-else, nested if-else, if-else-if ladder, break, continue, goto switch, nested switch, switch case and nested if.	4	3
	3.2	Loop control- for loops, nested for loops, while loops, do while loop.	3	3
	3.3	Array, initialization, array terminology, characteristics of an array, one dimensional array and operations	3	3
	3.4	Strings and standard functions, Introduction to pointers. Basics of a function, call by value & call by reference.	3	3
	3.5	Structure, Union, file handling.	2	3
4	**(Practical Session)**			
		1. Write a program to add two numbers 2. Write a program to swap two numbers with and without temporary variable 3. Write a program to calculate area and perimeter of circle 4. Write a program to find power of	30	4

		a number 5. Write a program to find out the largest among three numbers using nested if else 6. Write a program to check whether a character is a vowel or consonant 7. Write a program to implement Arithmetic calculator using switch. 8. Write a program to find sum of first N natural numbers 9. Write a program to implement Multiplication table of a number 10. Write a program to implement Check whether a number is prime or not using function 11. Write a program to implement Fibonacci series of a number 12. Write a program to print the following pattern 1 1 2 1 2 3 13. Write a program to print one dimensional array 14. Write a program to find the length of a string.(using function) 15. Program to implement pointer		
5	**(Teacher Specific Content)**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Use of ICT tools in conjunction with traditional classroom teaching methods ● Interactive sessions ● Class discussions
	MODE OF ASSESSMENT **Mode of Assessment**

Assessment Types	**A. Continuous Comprehensive Assessment (CCA)** **CCA for Practical (30 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva
	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination(70 Marks,3 hrs)** Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES

1. Kernighan, B. W., & Ritchie, D. M. (1988). The C programming language (2nd ed.). Prentice Hall.
2. Schildt, H. (2017). C: The complete reference (4th ed.). McGraw-Hill Education.

SUGGESTED READINGS

1. Kamthe, A. Programming in C (3rd ed.). Pearson Education.
2. P K Sinha & Priti Sinha - Computer Fundamentals , Fourth Edition, BPB Publications.

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<table>
<tr><td>Faculty/ Discipline</td><td colspan="5">Technology and Applied Sciences/ Software Development and System Administration</td></tr>
<tr><td>Programme</td><td colspan="5">B.Voc.(Honours) Software Development and System Administration</td></tr>
<tr><td>Course Name</td><td colspan="5">Mathematics for System and Software</td></tr>
<tr><td>Type of Course</td><td colspan="5">SDC</td></tr>
<tr><td>Course Code</td><td colspan="5">MG1SDCSDS102</td></tr>
<tr><td>Course Level</td><td colspan="5">100</td></tr>
<tr><td>Course Summary</td><td colspan="5">This course gives an introduction to basic concepts of Number Theory and Set Theory. Students will get a good understanding of matrix operations and basic concepts of determinants. This course offers an introductory session to Graph Theory. Students acquire knowledge to construct Boolean functions which can be applied in various logical circuits.</td></tr>
<tr><td>Semester</td><td>1</td><td colspan="2">Credits</td><td>4</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course Details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>4</td><td>0</td><td>0</td><td>60</td></tr>
<tr><td>Pre-requisites, if any</td><td colspan="5">NIL</td></tr>
</table>

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Apply the concept of Number Theory for solving congruence problems	A	1,2,3,10
2	Understand the basic concepts in Set Theory	U	1,2,3,10
3	Apply the Matrix Problem in various fields of science	A	1,2,3,10

4	Analyse graph theory concepts with real life problems	An	1,2,3,10
5	Analyse the truth tables and logical expressions associated with each logic gates	An	1,2,3,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**3**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 2	**3**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 3	**3**	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 4	**3**	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 5	**3**	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**2**

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Number Theory and Sets**			
	1.1	Division and division Algorithm	5	1
	1.2	Modular Arithmetic	5	1
	1.3	Set operations and set identities	5	2
Text1: Chapter 2 section 2, Chapter 3 section 4 (Relevant sections only. Proof of all theorems excluded)				
2	**Matrices and Determinants**			
	2.1	Definition and different types of Matrices	5	3

	2.2	Matrix operations and related Matrices	5	3
	2.3	Definition of Determinants. Solution of consistent non - homogeneous linear equations using Cramer's rule and Matrix inversion methods (only of order 2x2)	5	3
Text 2: Chapter 2 sections 2,4,5,6,9 (Relevant sections only, proof of all theorems excluded, problems of inverse must be of order 2x2)				
3	**Elementary Graph theory**			
	3.1	Definition, Graph terminologies, Special Graphs	5	4
	3.2	Euler Paths and Circuits	5	4
	3.3	Hamiltonian Paths and Circuits	5	4
Text 1: Chapter 9 sections 1,2,5 (Relevant sections only, proof of all theorems excluded, simple problems only)				
4	**Boolean Algebra**			
	4.1	Boolean Function	5	5
	4.2	SOP form of Boolean expression	5	5
	4.3	Logic Gates and representations	5	5
Text 1: Chapter10 sections 1,2,3 (Relevant sections only, proof of all theorems excluded, problems of inverse must be of order 2x2)				
5	**(Teacher Specific Contents)**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** Brainstorming Lecture, Explicit Teaching, Active Cooperative Learning
	MODE OF ASSESSMENT

<table>
<tr>
<td rowspan="2">Assessment Types</td>
<td>A. Continuous Comprehensive Assessment (CCA)

CCA for Theory : 30 Marks

<table>
<tr><th>Components</th><th>Mark Distribution</th></tr>
<tr><td>Test</td><td>15 marks</td></tr>
<tr><td>Assignment</td><td>10 marks</td></tr>
<tr><td>Viva voce</td><td>5 marks</td></tr>
</table>
</td>
</tr>
<tr>
<td>B. End Semester Evaluation (ESE) – Non-MCQ

ESE for Theory : Written Test(70 Marks , 2 hrs)

Part A: Very Short Answer Questions (5 out of 8 Questions)- (5*2=10 Marks)
Part B: Short Answer Questions (5 out of 8 Questions) - (5*6=30 Marks)
Part C:Essay Questions(2 out of 4 Questions) - (2*15=30 Marks)</td>
</tr>
</table>

REFERENCES

1. Rosen, Kenneth. H. Discrete Mathematics and its applications, 6th edition, McGraw Hill Publishing Co. New Delhi, 2006.
2. Grewal, B. S. Higher Engineering Mathematics,42thEdition, Khanna publications, 2012.

SUGGESTED READINGS

1. Kreyszig, Erwin. Advanced Engineering Mathematics, Wiley, India, 2006.

	Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/Software Development and System Administration**				
Programme	**B.Voc.(Honours) Software Development and System Administration**				
Course Name	**Digital Marketing & Techniques using AI**				
Type of Course	**MDC**				
Course Code	**MG1MDCSDS100**				
Course Level	**100**				
Course Summary	This course offers an in-depth understanding of digital marketing principles and the application of Artificial Intelligence (AI) in enhancing marketing effectiveness. It begins with core concepts such as inbound, content, email, and influencer marketing, highlighting the scope and importance of digital marketing. The course then explores how AI transforms marketing practices through tools for SEO, social media targeting, email automation, predictive analytics, and real-time bidding.				
Semester	1	**Credits**		3	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	0	0	45
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the nature, scope, and key concepts of digital marketing including content, inbound, email, and influencer marketing..	U	1,2,3,10

2	Understand how Artificial Intelligence enhances digital marketing practices such as SEO, social media targeting, automation, predictive analytics, and customer behavior analysis.	U	1,2,10
3	Apply marketing data using AI techniques for better decision-making. Understand the usage of various digital marketing tools for SEO, SEM, social media, content creation, analytics, and automation.	A	1,2,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	2	2	0	0	0	0	0	0	2
CO 2	3	3	0	0	0	0	0	0	0	3
CO 3	3	3	0	0	0	0	0	0	0	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I: Introduction to Digital Marketing *(10 Hours)***			
	1.1	Digital Marketing- Nature, Scope and Importance	3	1
	1.2	Core Concepts-Inbound Marketing, Content Marketing.	3	1
	1.3	Email Marketing, Influential Marketing	4	1
	Module II: AI in Digital Marketing *(15 Hours)*			

2	2.1	AI-Powered SEO and Content Generation	2	2
	2.2	AI in Social Media Marketing: Audience Targeting & Sentiment Analysis	2	2
	2.3	Email Marketing Automation with AI	2	2
	2.4	Predictive Customer Behavior & Lead Scoring	3	2
	2.5	Programmatic Advertising and Real-Time Bidding	3	2
	2.6	Role of Big Data and AI in Marketing Analytics	3	2
3	**MODULE III :Digital Marketing Tools and Automation(20 Hours)**			
	3.1	SEO (Search Engine Optimization) Tools - Google Search Console, SEMrush	2	3
	3.2	SEM & PPC Tools (Search Engine Marketing & Paid Ads) - Google Ads, Microsoft Advertising	2	3
	3.3	Email marketing Tools - Mailchimp, Active Campaign	2	3
	3.4	Social Media Marketing and Content Creation Tools - Hootsuite, Canva, Grammarly	2	3
	3.5	Analytic & Reporting tools - Google analytics, Marketing automation Tools – HubSpot	3	3
	3.6	Use of Chatbots and Virtual Assistants, Meta Ads Manager	3	3

	3.7	1. Use Canva (or similar) to design 3 promotional banners for a marketing campaign. 2. Create Google Ads. 3. Create Meta Ads.	6	3
4	**(Teacher Specific Content)**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Use of ICT tools in conjunction with traditional classroom teaching methods ● Interactive sessions ● Class discussions
Assessment Types	**MODE OF ASSESSMENT**
	A. Continuous Comprehensive Assessment (CCA) **CCA for Practical(25 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva
	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination (50 Marks, 1.5 hrs)** Part A : Short Answer Questions(5 out of 7 Questions) - (5*4=20 Marks) Part B : Practical Question (1 Question) – (1*20=20Marks) Record : 10 Marks

REFERENCES

1. Chaffey, D., & Ellis-Chadwick, F. (2019). Digital marketing (7th ed.). Pearson Education.

2. Ryan, D. (2016). Understanding digital marketing: Marketing strategies for engaging the digital generation(4th ed.). Kogan Page.
3. Kingsnorth, S. (2022). Digital marketing strategy: An integrated approach to online marketing (3rd ed.). Kogan Page.

SUGGESTED READINGS

1. Chaffey, D., & Ellis-Chadwick, F. (2019). Digital Marketing (7th ed.). Pearson.
2. Ryan, D. (2016). Understanding Digital Marketing: Marketing Strategies for Engaging the Digital Generation (4th ed.). Kogan Page.

SEMESTER – II

Semester: 2

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG2SDCSDS100	Computer System Design & Architecture	SDC	4	4	4	0	0
MG2SDCSDS101	Database Management Using MySQL	SDC	4	5	3	2	0
MG2SDCSDS102	PHP Programming	SDC	4	5	3	2	0
MG2MDCSDS100	Web Designing	MDC	3	4	2	2	0
MG2SDCSDS103	On-the-Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On the Job Training

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**				
Programme	**B.Voc.(Honours) Software Development and System Administration**				
Course Name	**Computer System Design & Architecture**				
Type of Course	**SDC**				
Course Code	**MG2SDCSDS100**				
Course Level	**100**				
Course Summary	This course introduces students to the internal structure and functioning of computer systems. It covers the design and organization of fundamental components such as the CPU, memory hierarchy, and input/output subsystems. The course enables students to apply theoretical knowledge to analyze and evaluate architectural choices in modern computer systems.				
Semester	2	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		4	0	0	60
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Recall and describe basic computer architecture components, instruction cycle, and performance metrics	K	1,2
2	Understand CPU organization, ALU design, and control unit functioning including pipelining	U	1,2

3	Understand knowledge of memory hierarchy, cache mapping, and virtual memory management	U	1,2,3,10
4	Analyze the integration of I/O subsystems and evaluate performance.	An	1,2,3,10
Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	0	0	0	0	0	0	0	0
CO 2	3	3	0	0	0	0	0	0	0	0
CO 3	3	3	2	0	0	0	0	0	0	1
CO 4	3	3	2	0	0	0	0	0	0	1

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I - Basics of Computer System Design (15 Hours)**			
	1.1	Introduction to computer systems: Functional units and data flow, Evolution of computer architecture: Von Neumann and Harvard models	4	1
	1.2	Instruction cycle: Fetch, Decode, Execute	3	1
	1.3	Addressing modes and instruction formats, Introduction to instruction sets: RISC vs CISC	4	1
	1.4	Performance metrics: Clock cycle, MIPS, CPI, throughput	4	1

2	**Module II - Central Processing Unit Design (15 Hours)**			
	2.1	CPU structure and operations, General register organization and stack organization	4	2
	2.2	Arithmetic and logic unit (ALU) design	3	2
	2.3	Control unit: Hardwired vs microprogrammed control	3	2
	2.4	Instruction pipelining and hazard resolution	2	2
	2.5	Superscalar and multithreaded architectures	3	2
3	**Module III - Memory Organization (15 Hours)**			
	3.1	Memory hierarchy: Register, Cache, RAM, ROM, Secondary storage	4	3
	3.2	Cache memory: Mapping techniques and replacement policies	3	3
	3.3	Virtual memory: Paging, segmentation, page replacement	4	3
	3.4	RAM types: SRAM, DRAM, ROM, PROM, Flash	4	3
4	**Module IV - Input/output and System Integration (15 Hours)**			
	4.1	I/O devices and their classifications	2	4
	4.2	I/O techniques: Programmed I/O, interrupt-driven I/O, DMA	4	4
	4.3	Bus structures and arbitration	3	4
	4.4	Interrupt handling and priority schemes	2	4
	4.5	Parallel/serial data transfer and system performance integration	4	4
5	**(Teacher Specific Content)**			

<table>
<tr><td>Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)

• Use of ICT tools in conjunction with traditional classroom teaching methods
• Interactive sessions
• Class discussions</td></tr>
<tr><td rowspan="3">Assessment Types</td><td>MODE OF ASSESSMENT

Mode of Assessment</td></tr>
<tr><td>A. Continuous Comprehensive Assessment (CCA)

CCA for Theory(30 Marks)

1. Written Test
2. Problem Based Assignment
3. Seminars</td></tr>
<tr><td>B. End Semester Evaluation (ESE) – Non -MCQ

ESE for Theory : Written Test(70 Marks , 2 hrs)

Part A: Very Short Answer Questions (10 out of 12 Questions) - (10*2=20 Marks)
Part B: Short Answer Questions (4 out of 6 Questions) - (4*5=20 Marks)
Part C: Essay Questions (2 out of 4 Questions) - (2*15=30 Marks)</td></tr>
</table>

REFERENCES

1. Tanenbaum, A. S., & Austin, T. (2012). Structured computer organization (6th ed.). Pearson Education.
2. Hamacher, C., Vranesic, Z., & Zaky, S. (2011). Computer organization (5th ed.). McGraw-Hill Education.
3. Hayes, J. P. (1998). Computer architecture and organization (3rd ed.). McGraw-Hill.

SUGGESTED READINGS

1. Patterson, D. A., & Hennessy, J. L. (2021). Computer organization and design: The hardware/software interface (6th ed.). Morgan Kaufmann.
2. Mano, M. M. (2017). Computer system architecture (3rd ed.). Pearson Education.
3. Stallings, W. (2021). Computer organization and architecture: Designing for performance (11th ed.). Pearson

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<table>
<tr><td>Faculty/ Discipline</td><td colspan="5">Technology and Applied Sciences/ Software Development and System Administration</td></tr>
<tr><td>Programme</td><td colspan="5">B.Voc.(Honours)Software Development and System Administration</td></tr>
<tr><td>Course Name</td><td colspan="5">Database Management Using MySQL</td></tr>
<tr><td>Type of Course</td><td colspan="5">SDC</td></tr>
<tr><td>Course Code</td><td colspan="5">MG2SDCSDS101</td></tr>
<tr><td>Course Level</td><td colspan="5">100</td></tr>
<tr><td>Course Summary</td><td colspan="5">This course offers a comprehensive understanding of database concepts and systems, focusing on relational databases and MySQL. It covers theoretical foundations, practical implementations, advanced topics, and real-world application. Students gain hands-on experience with SQL, MySQL, joins, subqueries, normalization, transaction management, and stored procedures.</td></tr>
<tr><td>Semester</td><td>2</td><td colspan="2">Credits</td><td>4</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course Details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>75</td></tr>
<tr><td>Pre-requisites, if any</td><td colspan="5">NIL</td></tr>
</table>

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Explain the architecture, models, and components of DBMS.	U	1,2,10

2	Design relational databases using ER modeling and normalization techniques.	A	1,2,3,10
3	Apply SQL commands to create and manipulate relational data.	A	1,2,3,4,10
4	Implement and query databases using MySQL. Understand and apply concepts of transaction management and indexing.	A	1,2,8,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	2	0	0	0	0	0	0	0	2
CO 2	3	3	2	0	0	0	0	0	0	2
CO 3	2	3	2	2	0	0	0	0	0	3
CO 4	3	3	0	0	0	0	0	2	0	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I: Introduction to DBMS and Data Modeling (15 Hours)**			
	1.1	Introduction to Database Systems- File System, DBMS, Advantages of DBMS, Components of DBMS, Database Users and DBA roles	4	1

	1.2	Database Architecture- 3-Level Architecture (External, Conceptual, Internal), Data Independence	4	1
	1.3	Data Models- Hierarchical, Network, Relational, Object-Oriented	3	1
	1.4	Entity-Relationship (ER) Model- Entities, Attributes, Relationships, ER Diagrams and Conventions, Mapping ER to Relational Schema	4	1
2	**Module II: Relational Database Design & Normalization (15 Hours)**			
	2.1	Relational Model-Concepts: Tuple, Attribute, Relation, Schema, Relational Integrity Constraints: Domain, Entity, Referential	3	2
	2.2	Relational Algebra (Overview of operations)	2	2
	2.3	Functional Dependencies	2	2
	2.4	Normalization-1NF, 2NF, 3NF, BCNF	3	2
	2.5	Keys: Primary, Candidate, Foreign Keys	3	2
	2.6	Introduction to Indexing and Views	2	2
	Module III: SQL and Transaction Management (15 Hours)			

3	**3.1**	Structured Query Language (SQL)-DDL: CREATE, ALTER, DROP, DML: INSERT, UPDATE, DELETE, DQL: SELECT with JOIN, GROUP BY, ORDER BY,DCL and TCL Commands (GRANT, REVOKE, COMMIT, ROLLBACK)	3	3
	3.2	Constraints: NOT NULL, UNIQUE, CHECK, DEFAULT	3	3
	3.3	Subqueries and Set Operations	3	3
	3.4	Transactions and Concurrency Control-ACID Properties, Isolation Levels, Deadlocks (Basic Concept)	3	3
	3.5	Basic Concepts of Database Security and Authorization	3	3
4	**Module IV: Practical – MySQL Programming Lab (30 Hours)**			
	4.1	MySQL Installation and Configuration, Creating and Managing Databases and Tables, SQL Query Execution-SELECT queries with filters and joins, Nested Queries and Aggregate Functions, String and Date Functions. Implementing Constraints and Keys, Views and Indexes in MySQL, Transactions and Locking Mechanisms	30	4

		Suggested Experiments: 1. Create a database with multiple tables and define relationships. 2. Insert, update, and delete data using SQL commands. 3. Implement joins and complex queries. 4. Create views and indexes.		
5	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Use of ICT tools in conjunction with traditional classroom teaching methods ● Interactive sessions ● Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment**
	A. Continuous Comprehensive Assessment (CCA) **CCA for Practical (30 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva

	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination(70 Marks,3 hrs)** Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES

1. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). Database system concepts (7th ed.). McGraw-Hill Education.
2. Elmasri, R., & Navathe, S. B. (2016). Fundamentals of database systems (7th ed.). Pearson.

SUGGESTED READINGS

1. Groff, J. R., & Weinberg, P. N. (2004). SQL: The complete reference (3rd ed.). McGraw-Hill/Osborne.
2. Tahaghoghi, S., & Williams, H. (2007). Learning MySQL. O'Reilly Media.

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Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development & System Administration**
Course Name	**PHP Programming**
Type of Course	**SDC**
Course Code	**MG2SDCSDS102**
Course Level	**100**
Course Summary	This course covers server-side web development using PHP to create dynamic, database-driven applications. It includes PHP fundamentals and advanced topics like form handling, sessions, and MySQL integration, with a focus on secure and efficient coding.

Semester	2	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	**Expected Course Outcome**	**Learning Domains**	**PO No**
1	Understand the fundamentals of PHP including syntax, data types, variables, constants, operators, control structures, and loops	U	1,2

2	Apply arrays and user-defined functions to structure PHP programs, including file inclusions.	A	2,3
3	Develop secure PHP-based forms using GET/POST methods, perform validation and sanitization, and manage sessions and cookies.	A	1,2,6,8
4	Perform CRUD operations using PHP-MySQL integration and dynamically display data on web pages.	A	1,2,9,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
CO 2	**0**	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
CO 3	**3**	**3**	**0**	**0**	**0**	**2**	**0**	**2**	**0**	**0**
CO 4	**3**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**2**	**1**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	**Units**	**Course description**	**Hr s**	**CO No.**
Upon completion of this course students will able to understand:				
1	**Module I - Introduction to PHP Fundamentals (15 Hours)**			
	1.1	Introduction to www, History, Understanding client/server roles, Apache, PHP, MySQL, XAMPP Installation.	3	1
	1.2	PHP Basic syntax, PHP data Types, PHP Variables, PHP Constants.	3	1
	1.3	PHP Expressions, PHP Operators, PHP Control Structures, PHP Loops. PHP Functions, include and require functions.	4	1
	1.4	PHP Enumerated Arrays, PHP Associative Arrays, Array Iteration, PHP Multi-Dimensional Arrays, Array Functions	5	1
2	**Module II - PHP Form Processing and Session Management (15 Hours)**			
	2.1	PHP Forms, Cookies & Sessions PHP Form handling.	5	2

	2.2	PHP GET, PHP POST, PHP Form Validation, PHP Form Sanitization.	5	2
	2.3	PHP Cookie handling, PHP Session Handling, PHP Login Session.	5	2
3	**Module III - PHP-MySQL Database Connectivity and Operations (15 Hours)**			
	3.1	PHP functions for MySQL connectivity and operation- mysql_connect, mysql_select_db, mysql_query.	5	3
	3.2	PHP functions for MySQL connectivity and operation mysql_fetch_row, mysql_fetch_array, mysql_result, mysql_list_fields, mysql_num_fields	5	3
	3.3	Insertion, updation and deletion of data using PHP, displaying data from MySQL in webpage.	5	3
4	***(Practical Session)***			
	Program to implement php variables, loops, foreach function, arrays and strings			
	4.1	1. Print sum of digits of a number. 2. Display elements in an array using foreach (). 3. Write a php script to perform array functions. 4. Display an array of Books using foreach () in Two-Dimensional Array. 5. Check if a given year is leap year or not using POST method. 6. Perform string functions. a. String Length b. Reverse of a string c. Words count	6	4
	Program to implement PHP Form ,Session Handling & database Connectivity			
	4.2	1. Design a website called 'EVENT MANAGEMENT SYSTEM'. Use PHP Form Handling & Session Handling. Create webpage for both Admin and User .Perform necessary database operations for them. A. Create a database called 'eventmanagement' B. Create a table called 'registration' with following fields(id,firstname,lastname,email,password,repeatpassword,contact,gender,username and usertype). C. Create a Script to connect database D. Create a webpage called 'registartion.html' E. Create a Script to insert data into registration table. F. Create a Script to display data from registration table. G. Create a Script to delete data from registration table. H. Perform login operation.	24	4
5	**(Teacher Specific Content)**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Use of ICT tools in conjunction with traditional classroom teaching methods ● Interactive sessions ● Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment**
	A. Continuous Comprehensive Assessment (CCA) **CCA for Practical (30 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva
	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination(70 Marks,3 hrs)** Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES

1. Nixon, R. (2021). Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5 (6th ed.). O'Reilly Media.
2. Tatroe, K., MacIntyre, P., & Lerdorf, R. (2013). Programming PHP (3rd ed.). O'Reilly Media.
3. Welling, L., & Thomson, L. (2016). PHP and MySQL Web Development (5th ed.). Addison-Wesley.

SUGGESTED READINGS

1. Nixon, R. (2021). Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5 (6th ed.). O'Reilly Media.
2. Ullman, L. (2017). PHP and MySQL for dynamic web sites: Visual quickpro guide (5th ed.). Peachpit Press.

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Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**Web Designing**
Type of Course	**MDC**
Course Code	**MG2MDCSDS100**
Course Level	**100**
Course Summary	This course helps students understand and apply the fundamentals of web design and interactivity. Students will learn HTML, CSS, basic JavaScript, and UI/UX principles. Through hands-on practice using modern tools like Canva or Figma, they will design and develop responsive, user-friendly websites.

<table>
<tr><td>Semester</td><td>2</td><td colspan="2">Credits</td><td>3</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course Details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>2</td><td>1</td><td>0</td><td>60</td></tr>
<tr><td>Pre-requisites, if any</td><td colspan="5">NIL</td></tr>
</table>

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the structure and design principles of web pages	U	1,2
2	Develop interactive and responsive web pages using HTML, CSS, and JavaScript	A	2,3,10
3	Develop functional and visually appealing website using modern web tools	A	2,3,10

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	0	0	0	0	0	0	0	0
CO 2	0	3	2	0	0	0	0	0	0	3
CO 3	0	3	3	0	0	0	0	0	0	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand				
1	**Module I - Basics of Web Design and Structure (15 Hours)**			
	1.1	Introduction to the web: browsers, servers, and the internet	3	1
	1.2	HTML elements and page structure	3	1
	1.3	Basic CSS styling: colors, fonts, spacing, layout (Flexbox/Grid)	3	1
	1.4	Typography and color theory in web design	3	1
	1.5	Introduction to design tools: Canva or Figma (basic mockup creation)	3	1
2	**Module II - Web Interactivity and User Experience (15 Hours)**			
	2.1	Introduction to JavaScript: variables, events	3	2
	2.2	Creating interactive elements: buttons, forms, image sliders	3	2

	2.3	Responsive design with media queries	3	2
	2.4	User experience (UX) basics: navigation, readability, mobile-first design	3	2
	2.5	Accessibility and usability tips	3	2
3	**Module III - Design for a Campus Event Management (30 Hours)**			
	3.1	1. Create a basic personal webpage using HTML and CSS 2. Design a homepage layout using Canva or Figma 3. Add a contact form with simple validation using JavaScript 4. Build a navigation menu with dropdown 5. Apply responsive design using media queries	30	3
4	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Use of ICT tools in conjunction with traditional classroom teaching methods ● Interactive sessions ● Class discussions
	MODE OF ASSESSMENT **Mode of Assessment**

<table>
<tr><td rowspan="2">Assessment Types</td><td>A. Continuous Comprehensive Assessment (CCA)

CCA for Practical (25 Marks)

1. Lab Assessment
2. Observation of Practical Skills
3. Viva</td></tr>
<tr><td>B. End Semester Evaluation (ESE)

ESE for Practical : Practical Examination(50 Marks,1.5 hrs)

Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks)
Part B : Practical Questions(1 Questions) – (1*20=20 Marks)
Record : 10 Marks</td></tr>
</table>

REFERENCES

1. Krug, S. (2014). Don't make me think, revisited: A common sense approach to web usability (3rd ed.). New Riders.
2. Robbins, J. N. (2018). Learning web design: A beginner's guide to HTML, CSS, JavaScript, and web graphics (5th ed.). O'Reilly Media.

SUGGESTED READINGS

1. Duckett, J. (2011). HTML and CSS: Design and build websites. Wiley.
2. Duckett, J. (2014). JavaScript and JQuery: Interactive front-end web development. Wiley.

SEMESTER – III

Semester: 3

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG3SDCSDS200	Software Engineering & Software Testing Principles	SDC	4	4	4	0	0
MG3SDCSDS201	Data structure using C++	SDC	4	5	3	2	0
MG3SDCSDS202	Desktop Application Development using .NET & SQL Server	SDC	4	5	3	2	0
MG3MPCSDS200	Database Management System (DBMS)	MPC	4	5	3	2	0
MG3MDCSDS200	E-governance and Digital Kerala System	MDC	3	3	3	0	0
MG3SDCSDS203	On-the-Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On the Job Training

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<table>
<tr><td>Faculty/ Discipline</td><td colspan="5">Technology and Applied Sciences/ Software Development and System Administration</td></tr>
<tr><td>Programme</td><td colspan="5">B.Voc.(Honours) Software Development and System Administration</td></tr>
<tr><td>Course Name</td><td colspan="5">Software Engineering & Software Testing Principles</td></tr>
<tr><td>Type of Course</td><td colspan="5">SDC</td></tr>
<tr><td>Course Code</td><td colspan="5">MG3SDCSDS200</td></tr>
<tr><td>Course Level</td><td colspan="5">200</td></tr>
<tr><td>Course Summary</td><td colspan="5">This course provides a comprehensive overview of Software Engineering and Testing, covering software development life cycle models, project management. It equips students knowledge of software testing types, techniques, and tools—including manual and automated testing—alongside quality assurance practices and maintenance strategies.</td></tr>
<tr><td>Semester</td><td>3</td><td colspan="2">Credits</td><td>4</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course Details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>4</td><td>0</td><td>0</td><td>60</td></tr>
<tr><td>Pre-requisites, if any</td><td colspan="5">NIL</td></tr>
</table>

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand fundamental concepts of software engineering, SDLC, software process models, software requirement analysis and software design software project management.	U	1,2,3,10

2	Apply software development life cycle (SDLC) and software testing life cycle (STLC) principles to design, plan, understand levels of testing and testing strategies for different project contexts.	A	1,2,3,10
3	Analyze and apply appropriate manual and automated testing techniques, strategies, and tools to ensure software quality across functional, structural, and performance aspects, with a clear understanding of white box and black box testing methods.	An	1,2,8,10
4	Demonstrate advanced testing strategies and explain the role of software quality assurance in delivering reliable software.	A	1,2,3,8
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**3**	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**3**
CO 2	**3**	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**3**
CO 3	**3**	**3**	**0**	**0**	**0**	**0**	**0**	**2**	**0**	**3**
CO 4	**3**	**3**	**2**	**0**	**0**	**0**	**0**	**2**	**0**	**0**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
	Module I - Software Engineering Fundamentals (20 hours)			
1	**1.1**	**Software Engineering –** Introduction-Definition, Need, Scope, Characteristics of software, Challenges in software development -Software Process and Lifecycle Software Development Life Cycle (SDLC) – Phases Process framework and activities	5	1

	1.2	**Software Process Models** Waterfall Model Incremental Model Spiral Model V-Model Agile methodologies (Scrum, XP) – overview	5	1
	1.3	**Software Requirement Analysis:** Requirement Elicitation – Use Case, DFD, Data Dictionaries. **Software Design:** Definition, Various types, Modularity.	5	1
	1.4	**Software Project Management Basics** Project Planning and Scheduling Cost Estimation techniques: LOC, FP Risk Management: Risk identification, analysis.	5	1
2	**Module II - Software Testing Fundamentals (10 hours)**			
	2.1	**Basics of Software Testing-** Definition, objectives, principles, Difference between verification and validation-Software testing life cycle (STLC).	3	2
	2.2	**Testing Strategies** Static Testing (Code reviews, inspections) Dynamic Testing (test execution)-Test Case Design Techniques.	3	2
	2.3	**Levels of testing**: Unit testing, integration testing, system testing, validation testing, Alpha, beta and acceptance testing, functional testing.	4	2
3	**Module III - Software Testing Techniques and Automation (15 hours)**			
	3.1	**Testing Techniques in Detail** Unit Testing, Integration Testing, System Testing, Acceptance Testing Testing, Acceptance Testing.	**5**	3

	3.2	**Black Box Testing Techniques:** Equivalence Partitioning, Boundary Value Analysis. **White Box Testing Techniques:** Statement Coverage, Decision Coverage, Path Testing, Flow Graph Testing Approaches.	5	3
	3.3	**Automation in Testing :** Definition and need for automation, Benefits and limitations of automated testing, Manual vs automated testing.	5	3
4	**Module IV - Advanced Software Testing and Quality Assurance (15 Hours)**			
	4.1	**Advanced Testing Strategies** Object-Oriented Testing – class-level testing, method testing Web Application Testing usability, security, compatibility testing Regression Testing – tools and techniques Performance Testing – load, stress, scalability Test Defect Management-Defect Life Cycle Bug reporting and tracking tools.	5	4
	4.2	**Automation Tools (Overview)** Selenium – features and usage JUnit – unit test framework in Java Introduction to test scripts. Software Quality Assurance (SQA) **-**SQA activities and goals SQA Standards: ISO 9001, CMMI levels Reviews and audits.	5	
	4.3	**Software Maintenance and Reengineering**-Types of Maintenance: Corrective, Adaptive, Perfective, Preventive Legacy system challenges Introduction to software reengineering.	5	
5	**(Teacher Specific Content)**			

<table>
<tr><td>Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)
Use of ICT tools in conjunction with traditional classroom teaching methods
Interactive sessions
Class discussions</td></tr>
<tr><td rowspan="3">Assessment Types</td><td>MODE OF ASSESSMENT
Mode of Assessment</td></tr>
<tr><td>A. Continuous Comprehensive Assessment (CCA)
CCA for Theory(30 Marks)
1. Written Test
2. Problem Based Assignment
3. Seminars</td></tr>
<tr><td>B. End Semester Evaluation (ESE) – Non-MCQ
ESE for Theory : Written Test(70 Marks , 2 hrs)
Part A: Very Short Answer Questions (10 out of 12 Questions)- (10*2=20Marks)
Part B: Short Answer Questions (4 out of 6 Questions) - (4*5=20 Marks)
Part C: Essay Questions (2 out of 4 Questions) - (2*15=30 Marks)</td></tr>
</table>

REFERENCES:

1. Sommerville, Ian. Software Engineering (10th ed.), Pearson
2. Jalote, Pankaj. An Integrated Approach to Software Engineering, Narosa Publishing

SUGGESTED READING:

1. Pressman, R. S., & Maxim, B. R. Software Engineering: A Practitioner's Approach (9th ed.), McGraw-Hill.
2. Sommerville, I. (2016). Software engineering (10th ed.). Pearson Education.
3. Jalote, P. (2012). An integrated approach to software engineering (4th ed.). Springer.

Mahatma Gandhi University Kottayam

<table>
<tr><td>Faculty/ Discipline</td><td colspan="5">Technology and Applied Sciences/ Software Development and System Administration</td></tr>
<tr><td>Programme</td><td colspan="5">B.Voc.(Honours) Software development and system administration</td></tr>
<tr><td>Course Name</td><td colspan="5">Data Structure using C++</td></tr>
<tr><td>Type of Course</td><td colspan="5">SDC</td></tr>
<tr><td>Course Code</td><td colspan="5">MG3SDCSDS201</td></tr>
<tr><td>Course Level</td><td colspan="5">200</td></tr>
<tr><td>Course Summary</td><td colspan="5">Introduces core data structures using C++, including arrays, stacks, queues, linked lists, trees, and file systems, with emphasis on both static and dynamic implementations. Covers key operations and applications such as memory allocation, sparse matrices, expression evaluation, and tree traversal. Includes searching and sorting algorithms, as well as file organization and hashing techniques for efficient data storage and retrieval.</td></tr>
<tr><td>Semester</td><td>3</td><td colspan="2">Credits</td><td>4</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course Details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>75</td></tr>
<tr><td>Pre-requisites, if Any</td><td colspan="5">NIL</td></tr>
</table>

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the basic structure of data representation using arrays and apply searching and sorting algorithms to solve computational problems.	U	1,2,3,10

2	Implement stack and queue structures and apply them in algorithmic solutions involving expression conversion, evaluation, and memory management.	A	1,2,10
3	Develop and manipulate dynamic data structures using linked lists to implement linear data types and understand basic memory management techniques.	An	1,2,10
4	Implement and traverse tree structures and understand file organization techniques for efficient data storage and retrieval.	A	1,2,3,10
****Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)***			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**2**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 2	**2**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 3	**2**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 4	**2**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**2**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	**Units**	**Course description**	**Hrs**	**CO No.**
Upon completion of this course students will able to understand:				
	Module I – Data Structure Fundamentals (17 Hours)			
	1.1	Concept of Structured data - Data structure definition, Different types and classification of data structures, Arrays	5	1

1	**1.2**	Memory allocation and implementation of arrays in memory, array operations, Applications - sparse matrix representation and operations,	6	1
	1.3	polynomials representation and addition, Concept of search and sort – linear search, binary search, selection sort, insertion sort, quick sort.	6	1
2	**Module II – Stacks, Queue and Linked List Implementation (17 Hours)**			
	2.1	Stacks – Concepts, organization and operations on stacks using arrays (static), examples, Applications Conversion of infix to postfix and infix to prefix, postfix evaluation, subprogram calls and execution, Multiple stacks representation.	5	2
	2.2	Queues - Concepts, organization and operations on queues, examples. Circular queue – limitations of linear queue, organization and operations on circular queue. Double ended queue, Priority queue.	4	2
	2.3	Linked list: Concept of dynamic data structures, linked list, types of linked list, linked list using pointers, insertion and deletion examples, circular linked list, doubly linked lists Applications. linked stacks and queues, memory management basic concepts, garbage collection.	8	2
3	**Module III – Trees and Files (11 Hours)**			
	3.1	Trees - Concept of recursion, trees, tree terminology, binary trees, representation of binary trees, strictly binary trees, complete binary tree, extended binary trees,	3	2

	3.2	creation and operations on binary tree, binary search trees, Creation of binary search tree, tree traversing methods – examples, binary tree representation of expressions	4	3
	3.3	. File - Definition, Operations on file (sequential), File organizations - sequential, Indexed sequential, random files, linked organization, inverted files, cellular partitioning, hashing – hash tables, hashing functions, collisions, collision resolving methods.	4	3
4	**Module – IV (Practical sections)**			
	4.1	Implement array insertion, Deletion Implement Linear Search, Binary Search, Bubble Sort, Selection Sort, Merge Sort	10	4
	4.2	Implement stack and Queue operations using arrays Implement Circular Queue operations.	10	
	4.3	Implement operations on a linked list.	10	
5	**(Teacher Specific Content)**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Use of ICT tools in conjunction with traditional classroom teaching methods ● Interactive sessions ● Class discussions
	MODE OF ASSESSMENT **Mode of Assessment**

<table>
<tr>
<td rowspan="2">Assessment Types</td>
<td>A. Continuous Comprehensive Assessment (CCA)

CCA for Practical (30 Marks)

1. Lab Assessment
2. Observation of Practical Skills
3. Viva</td>
</tr>
<tr>
<td>B. End Semester Evaluation (ESE)

ESE for Practical : Practical Examination(70 Marks,3 hrs)

Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks)
Part B : Practical Questions(2 Questions) – (2*15=30 Marks)
Viva : 10 Marks
Record : 10 Marks</td>
</tr>
</table>

REFERENCES

1. G.S Baluja (2004). Data Structures Through C (A Practical Approach) (2nd Edition).Danapat Rai & Co.

2. Ellis Horowitz and Sartaj Sahni. Fundamentals of Data Structures (2nd Edition). Galgotia Publications.

SUGGESTED READINGS

1. Seymour Lipschutz, Theory and Problems of Data Structures, Schaum's Outline Series,2006, McGraw Hill

2. Yedidyah Lanngsam, Moshe Augustein, Aaron M Tenenbaum- Data structures using C and C++, Second Edition, Prentice Hall

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**Desktop Application Development using .NET & SQL Server**
Type of Course	**SDC**
Course Code	**MG3SDCSDS202**
Course Level	**200**
Course Summary	This course teaches students how to design, develop, and deploy desktop-based applications using Microsoft.NET Framework and SQL Server. It covers UI design, data binding, event-driven programming, form validation, ADO.NET, and database operations.

Semester	3	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the architecture and components of the .NET Framework.	U	1,2,3,10
2	Develop basic applications using C# and Windows Forms	A	1,2,3,4,10

3	Use SQL to manage relational databases and write queries. Integrate .NET applications with SQL Server using ADO.NET.	A	1,2,3,10
4	Integrate .NET applications with SQL Server using ADO.NET.	C	1,2,3,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	1	3	1	0	0	0	0	0	0	2
CO 2	1	3	1	1	0	0	0	0	0	1
CO 3	1	3	1	0	0	0	0	0	0	2
CO 4	1	3	2	0	0	0	0	0	0	1

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I - Introduction to .NET and C# Programming (15 Hours)**			
	1.1	Overview of the .NET Framework- CLR, CTS, CLS, BCL,Assemblies, Namespaces	5	1
	1.2	Introduction to C# Programming- Data Types, Variables, Operators, Control Structures (if, switch, loops), Arrays and Methods, Exception Handling	5	1
	1.3	Object-Oriented Programming Concepts- Classes and Objects,Inheritance, Polymorphism, Interfaces	5	1

2	**Module II - Windows Forms Application Development (15 Hours)**			
	2.1	Visual Studio IDE Overview	2	2
	2.2	Windows Forms and Controls-Buttons, TextBoxes, ComboBoxes, Labels, Events and Event Handling, Form Validation and ToolTips	5	2
	2.3	File Handling in C#	3	2
	2.4	Multi-document Interface (MDI) Applications	3	2
	2.5	Error Providers and Dialog Controls	2	2
3	**Module III - SQL and ADO.NET Integration (15 Hours)**			
	3.1	Introduction to SQL Server- Creating Tables, DDL & DML Operations, Joins, Constraints, Indexes, Views and Stored Procedures	7	3
	3.2	Introduction to ADO.NET- Connection, Command, DataReader, DataAdapter, DataSet, Data Binding to Controls (e.g., DataGridView), Executing Queries from .NET, CRUD Operations with Parameterized Queries	8	3
4		**(Practical Session)**		
	4.1	***Sample Exercises:*** 1. Create a simple C# console and Windows Forms application. 2. Design and validate a login form using Windows Forms. 3. Create a SQL Server database with student/employee records. 4. Perform CRUD operations in SQL and connect the database to C# using ADO.NET.	**30**	**4**

		5. Develop a mini desktop application (e.g., Library/Inventory System).		
5	**(Teacher Specific Content)**			

Assessment Types	**A. Continuous Comprehensive Assessment (CCA)** **CCA for Practical (30 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva
	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination(70 Marks,3 hrs)** Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES

1. Troelsen, A., & Japikse, P. (2020). Pro C# 8 with .NET Core 3. Apress.
2. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). Database system concepts (7th ed.). McGraw-Hill.

SUGGESTED READINGS

1. Beaulieu, A. (2009). Learning SQL. O’Reilly.
2. Hamilton, B. (2003). ADO.NET in a nutshell. O’Reilly.

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Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**				
Programme	**B.Voc.(Honours) Software Development and System Administration**				
Course Name	**Database Management System (DBMS)**				
Type of Course	**MPC**				
Course Code	**MG3MPCSDS200**				
Course Level	**200**				
Course Summary	This course introduces the fundamental concepts of database systems with a focus on data models, relational database design, SQL programming, and practical implementation using a relational DBMS. It covers theoretical foundations and hands- on practices to equip students with the skills required to design, implement, and manage database applications efficiently.				
Semester	3	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre- requisites, if any	**NIL**				

COUURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the fundamental concepts of databases and database management systems.	U	1,2,3,10
2	Learn data models, relational algebra, and normalization techniques	An	1,2,3,10
3	Gain knowledge in SQL for data manipulation and database queries	A	1,2,3,9,10
4	Develop skills in designing and implementing relational databases.	A	1,2,3,9,10
****Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)***			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**2**	**2**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 2	**3**	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 3	**2**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**1**	**3**
CO 4	**3**	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**1**	**3**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I: Introduction to Databases and Data Models (15 Hours)**			
	1.1	Overview of databases: File system vs. DBMS	3	1
	1.2	Characteristics and advantages of DBMS	3	1

	1.3	Types of DBMS (Hierarchical, Network, Relational, NoSQL – brief	3	1
	1.4	Data models: Hierarchical, Network, Relational	3	1
	1.5	Entity-Relationship (ER) Model: Entities, attributes, relationships, ER diagrams	3	1
2	**Module II: Relational Database Design and Normalization (15 Hours)**			
	2.1	Relational model concepts: tables, keys (primary, foreign, candidate)	4	2
	2.2	Relational algebra: selection, projection, union, join, Cartesian Product	4	2
	2.3	Functional dependencies	3	2
	2.4	Normalization: 1NF, 2NF, 3NF, BCNF	4	2
3	**Module III: SQL and Transaction Management (15 Hours)**			
	3.1	SQL basics: DDL, DML, DCL	3	3
	3.2	SQL Queries: SELECT, WHERE, JOINs, GROUP BY, ORDER BY	5	3
	3.3	Subqueries, Views, Indexes	3	3
	3.4	Transactions: ACID properties	4	3
4	**4.1**	**(Practical Session)** 1. Install and set up DBMS software 2. Create databases and tables using SQL 3. Write SQL queries for: Data insertion, update,	30	4

		deletion, Table joins and subqueries, Views and indexes		
5	**(Teacher Specific Content)**			

<table>
<tr><td rowspan="1">Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)
● Use of ICT tools in conjunction with traditional classroom teaching methods
● Interactive sessions
● Class discussions</td></tr>
<tr><td rowspan="3">Assessment Types</td><td>MODE OF ASSESSMENT
Mode of Assessment</td></tr>
<tr><td>A. Continuous Comprehensive Assessment (CCA)
CCA for Practical (30 Marks)
1. Lab Assessment
2. Observation of Practical Skills
3. Viva</td></tr>
<tr><td>B. End Semester Evaluation (ESE)
ESE for Practical : Practical Examination(70 Marks,3 hrs)
Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks)
Part B : Practical Questions(2 Questions) – (2*15=30 Marks)
Viva : 10 Marks
Record : 10 Marks</td></tr>
</table>

REFERENCES

1. Elmasri, R., & Navathe, S. B. (2016). Fundamentals of database systems (7th ed.). Pearson.
2. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2020). Database system concepts (7th ed.). McGraw-Hill Education.

SUGGESTED READINGS

1. Ramakrishnan, R., & Gehrke, J. (2003). Database management systems (3rd ed.).
2. Stephens, R., & Plew, R. R. (2008). Sams teach yourself SQL in 10 minutes (3rd ed.). Sams Publishing.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**				
Programme	**B.Voc.(Honours) Software Development and System Administration**				
Course Name	**E-Governance and Digital Kerala System**				
Type of Course	**MDC**				
Course Code	**MG3MDCSDS200**				
Course Level	**200**				
Course Summary	This course introduces the principles, architecture, and implementation of e-Governance systems with a focus on initiatives in Kerala. It emphasizes the technological, legal, and administrative frameworks necessary for building inclusive, transparent, and efficient digital governance.				
Semester	3	**Credits**		3	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	0	0	45
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the concept and components of e-Governance and evaluate its role in improving public service delivery.	U	1,2,3,6,7,10

2	Analyze the technical and legal frameworks that support secure and scalable e-Governance platforms.	An	1,2,3,6,8,10
3	Examine the implementation of e-Governance in Kerala and evaluate its impact on citizen empowerment and public service transparency.	A	1,2,3,6,8,9,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**2**	**2**	**2**	**0**	**0**	**3**	**2**	**0**	**0**	**2**
CO 2	**3**	**3**	**2**	**0**	**0**	**2**	**0**	**2**	**0**	**2**
CO 3	**3**	**2**	**2**	**0**	**0**	**3**	**0**	**2**	**2**	**2**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I – Introduction to e-governance (15 Hours)**			
	1.1	Definition and objectives of e-Governance, Evolution of governance: From manual to digital, Types of e-Governance (G2G, G2C, G2B, G2E)	7	1
	1.2	Benefits and challenges of e-Governance, Components of e-Governance: Infrastructure, connectivity, services, National e-Governance Plan (NeGP), Digital India Programme.	8	1
	Module II – e-governance Systems and Security (15 Hours)			

2	**2.1**	E-Governance Architecture: Front-end and back-end systems, ICT tools in governance: Portals, mobile apps, databases, E-Governance standards and interoperability	7	2
	2.2	Cybersecurity in governance: Data privacy, authentication (Aadhaar), encryption, Digital identity management and trust systems, Legal framework: IT Act 2000, Right to Information (RTI), Data Protection	8	2
3	**Module III - Digital Governance Initiatives in Kerala (15 Hours)**			
	3.1	Overview of Digital Kerala Mission, Key initiatives: eDistrict, Akshaya, eHealth, Kerala Fibre Optic Network (KFON), FRIENDS, Ente Nadappu, Smart governance and local self-government digitalization	7	3
	3.2	Role of Akshaya Centres in citizen empowerment, Inclusivity and gender in digital governance.	4	3
	3.3	Case studies of successful e-Governance projects in Kerala (Practicum Content)	4	3
4	**(Teacher specific Content)**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Use of ICT tools in conjunction with traditional classroom teaching methods ● Interactive sessions ● Class discussions
	MODE OF ASSESSMENT **Mode of Assessment**

Assessment Types	**A. Continuous Comprehensive Assessment (CCA)** **CCA for Theory(25 Marks)** 1. Written Test 2. Assignment 3. Seminar 4. Case Study Report Submission
	B. End Semester Evaluation (ESE) – Non-MCQ **ESE for Theory : Written Test(50 Marks,1.5 hrs)** Part A : Very Short Answer Questions (10 out of 12 Questions) - (10*2=20 Marks) Part B : Short Answer Questions (4 out of 6 Questions) - (4*5=20 Marks) Part C : Essay Questions (1 out of 3 Questions) - (1*10=10 Marks)

REFERENCES

1. Bhatnagar, S. C. (2009). Unlocking E-Government Potential: Concepts, Cases and Practical Insights. SAGE Publications.
2. Heeks, R. (2006). Implementing and Managing eGovernment: An International Text. SAGE Publications.

SUGGESTED READINGS

1. Pankaj Sharma – E-Governance: The New Age Governance
2. A.P. Singh & M.P. Gupta – Government Online: Opportunities and Challenges
3. Ministry of Electronics & IT (MeitY), Govt. of India – NeGP & Digital India policy document
4. Kerala State IT Mission – official reports, mission documents (Digital Kerala)
5. Ashok Agarwal – E-Governance: Case Studies

SEMESTER – IV

Semester: 4

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG4SDCSDS200	Artificial Intelligence and Machine Learning	SDC	4	4	4	0	0
MG4SDCSDS201	Python Programming	SDC	4	5	3	2	0
MG4SDCSDS202	Operating System & Linux Administration	SDC	4	5	3	2	0
MG4MPCSDS200	.NET & SQL Server - Desktop Application Development	MPC	4	5	3	2	0
MG4SECSDS200	MIS Deployment in IT	SEC	3	3	3	0	0
MG4VACSDS200	Social Impact of Technologies	VAC	3	3	3	0	0
MG4INTSDS200	Internship	INT	2				

L — Lecture, P — Practical/Pracicum , O — On the Job Training

	Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**Artificial Intelligence and Machine Learning**
Type of Course	**SDC**
Course Code	**MG4SDCSDS200**
Course Level	**200**
Course Summary	This course introduces students to the foundational concepts, techniques, and tools of Artificial Intelligence (AI) and Machine Learning (ML). It covers key theoretical principles of AI, problem-solving methods, machine learning pipelines, algorithms, model evaluation techniques, and ethical considerations. Students will understand the interconnections between AI, ML, Deep Learning, and Data Science, explore real-world applications, and be introduced to standard ML tools and platforms.

Semester	4	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		4	0	0	60
Pre-requisites, if Any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the foundational concepts and historical development of Artificial Intelligence and its applications.	U	1,10

2	Apply machine learning types, pipelines, and algorithms to solve basic data- driven problems.	A	2,10
3	Analyze and interpret performance metrics and model behaviour using evaluation techniques.	An	1.2,10
4	Apply and analyze simple AI/ML model workflows using conceptual tools and examine responsible AI practices.	A	3,8,10
Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 2	**0**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 3	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
CO 4	**0**	**0**	**2**	**0**	**0**	**0**	**0**	**3**	**0**	**1**

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COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
	Module I - Introduction to Artificial Intelligence (15 Hours)			
1	**1.1**	Definition and Scope of AI, History and Evolution of AI	3	1
	1.2	Types of AI: Narrow, General, and Super AI, Applications in real life: healthcare, security, education, business	3	1
	1.3	Intelligent Agents: structure and types, Environments: deterministic/stochastic, episodic/sequential	4	1
	1.4	Problem Solving: state space search, uninformed search (BFS, DFS)	3	1

	1.5	Heuristic Search	2	1
	Module II - Fundamentals of Machine Learning (15 Hours)			
2	**2.1**	Relationship between AI, ML, Deep Learning, and Data Science	2	2
	2.2	Types of ML: Supervised, Unsupervised, Reinforcement Learning	2	2
	2.3	ML Pipeline: Data collection, Pre-processing, Model Building, Evaluation Datasets and Features	4	2
	2.4	Overview of Algorithms: Supervised: Linear Regression, Logistic Regression Unsupervised: K-means, Hierarchical Clustering	4	2
	2.5	Python libraries: scikit-learn, pandas, NumPy	3	2
	Module III - Model Evaluation and Performance Analysis (15 Hours)			
3	**3.1**	Data Splitting: Training, Testing, Validation Overfitting vs Under fitting	3	3
	3.2	Performance Metrics for Classification: Confusion Matrix, Accuracy, Precision, Recall	4	3
	3.3	Performance Metrics for Regression: Mean Squared Error (MSE), R^2 Score	4	3
	3.4	Cross Validation Concepts, Case Study Discussion (e.g., Email Spam Detection, Medical Diagnosis)	4	3
	Module IV - AI/ML Applications, Tools, and Ethics (15 Hours)			
	4.1	Introduction to Machine Learning Tools :Jupyter Notebook, Google Colab,	4	4
	4.2	Python-based tools: scikit-learn, TensorFlow	2	4

4	**4.3**	Steps in Building a Simple Model, Real-world Applications: Chatbots, Sentiment Analysis, Recommender Systems	4	4
	4.4	Ethical Aspects of AI: Bias, Privacy, Transparency, Job Displacement	3	4
	4.5	Principles of Responsible AI	2	4
5	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment**
	A. Continuous Comprehensive Assessment (CCA) **CCA for Theory(30 Marks)** 1. Written Test 2. Seminars 3. Assignments
	B. End Semester Evaluation (ESE) – Non - MCQ **ESE for Theory : Written Test (70 Marks , 2 hrs)** Part A: Very Short Answer Questions (10 out of 12 Questions) - (10*2=20 Marks) Part B: Short Answer Questions (4 out of 6 Questions) - (4*5=20 Marks) Part C: Essay Questions (2 out of 4 Questions) - (2*15=30 Marks)

REFERENCES

1. Mitchell, T. M. (1997). Machine Learning. McGraw-Hill.
2. Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning.
3. MIT Press. Kelleher, J. D., Mac Carthy, M., & Korvir, B. (2018). Fundamentals of Machine Learning

for Predictive Data Analytics: Algorithms, Worked Examples, and Case Studies (2nd ed.). MIT Press.

SUGGESTED READINGS

1. Russell, S. J., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach (4th ed.). Pearson Education.
2. Géron, A. (2019). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems (2nd ed.). O'Reilly Media.
3. Shalev-Shwartz, S., & Ben-David, S. (2014). Understanding Machine Learning: From Theory to Algorithms. Cambridge University Press.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**Python Programming**
Type of Course	**SDC**
Course Code	**MG4SDCSDS201**
Course Level	**200**
Course Summary	Understand the basic concepts, syntax, and structures of Python. Apply control structures, functions, and modules to solve programming tasks. Work with Python data structures and perform file operations. Create and test practical Python applications through hands-on programming.

Semester	4	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if Any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the basics of Python and its evolution, Apply appropriate data types and operations for simple problems, Demonstrate input-output handling in Python programs.	U	1,2,3,10
2	Use decision-making and iteration to solve programming, Problem Develop modular Python programs using functions, Utilize modules and built-in libraries to enhance functionality.	A	1,2,3,10

3	Implement Python data structures to store and process data, manipulate string data effectively, perform file operations using Python.	A	1,2,3,4,9,10
4	Develop and execute Python programs for real-world problems, integrate data structures and functions in solutions, build a simple project demonstrating all key concepts.	C	1,2,3,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**2**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 2	**2**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 3	**2**	**3**	**1**	**1**	**0**	**0**	**0**	**0**	**1**	**3**
CO 4	**2**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**2**

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COURSE CONTENT

Content for Classroom transaction (Units)

Module	**Units**	**Course description**	**Hrs**	**CO No.**
Upon completion of this course students will able to understand:				
1	**Module I - Introduction to Python Programming (15 Hours)**			
	1.1	Introduction to Python: History, features, installation, IDEs Python syntax, keywords, identifiers, variables, comments	5	1

	1.2	Data types: int, float, str, bool, None Type conversion and casting	5	1
	1.3	Input and Output functions Operators and Expressions	5	1
2	**Module II - Control Structures and Functions in Python (16 Hours)**			
	2.1	Conditional statements: if, else-if. Loops: for, while, break, continue, pass Modules and Packages- importing and using Operators and Expressions	5	2
	2.2	Functions: definition, parameters, return statement, Variable scope, recursion, anonymous functions (lambda)	5	2
	2.3	Modules and Packages – importing and using Operators and Expressions	6	2
3	**Module III - Data Structures and File Handling in Python (14 Hours)**			
	3.1	Lists, Tuples, Sets, Dictionaries – creation, methods, operations, Comprehensions – list, dict, set	7	3
	3.2	String operations and methods, Iterating over data structures- Introduction to file handling – open, read, write, append, close	7	3
4	**Practical Session**			
	4.1	1. Display “Hello, World!” 2. Swap two numbers using a temporary variable 3. Convert Celsius to Fahrenheit 4. Find the square root of a number 5. Calculate area and perimeter of a circle 6. Find largest of three numbers 7. Check whether a number	30	4

		is even or odd 8. Check if a year is a leap year 9. Print Fibonacci series up to n terms 10. Check if a number is prime 11. Find factorial of a number using recursion 12. Function to count vowels in a string 13. Lambda function to square all elements in a list 14. Program using built-in module (e.g., math, random) 15. Recursive function to compute power of a Number		
5	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment** **A. Continuous Comprehensive Assessment (CCA)** **CCA for Practical (30 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva

	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination(70 Marks,3 hrs)** Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions (2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES

1. Think Python: How to Think Like a Computer Scientist, Author: Allen B. Downey, Publisher: O'Reilly Media, Edition: 2nd Edition
2. Python Programming: An Introduction to Computer Science, Author: John Zelle Publisher: Franklin, Beedle & Associates
3. Programming in Python 3, Author: Mark Summerfield, Publisher: Pearson Education

SUGGESTED READINGS

1. Learning Python Author: Mark Lutz, Publisher: O'Reilly Media
2. Introduction to Computing and Problem Solving with Python Authors: Jeeva Jose, P. Sojan Lal, Publisher: Khanna Publishers

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**				
Programme	**B.Voc.(Honours) Software Development and System Administration**				
Course Name	**Operating system & Linux System Administration**				
Type of Course	**SDC**				
Course Code	**MG4SDCSDS202**				
Course Level	**200**				
Course Summary	This course offers an in-depth overview of operating systems, emphasizing their architecture, functions, and key components. It covers essential topics such as process handling, scheduling techniques, synchronization methods, memory management, and approaches for managing deadlocks and virtual memory. Gain hands-on experience with Linux commands, shell scripting, and file system management.				
Semester	4	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the basic structure, functions, services, and types of system calls in an operating system.	U	1,2,10
2	Analyze and apply scheduling algorithms and process coordination techniques like semaphores and monitors.	An	1,2

3	Evaluate deadlock-handling strategies including prevention, avoidance, detection, and memory management techniques.	E	1,2
4	Gain hands-on experience with Linux commands, shell scripting, and file system management.	A	1,2,4,9,10
Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	0	0	0	0	0	0	0	2
CO 2	3	3	0	0	0	0	0	0	0	0
CO 3	3	3	0	0	0	0	0	0	0	0
CO 4	3	3	0	2	0	0	0	0	3	3

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COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I - Introduction to OS (11 Hours)**			
	1.1	OS Definition, Functions, Evolution of OS.	2	1
	1.2	OS Structure Operating System Operations.	2	1
	1.3	Operating System Services, User Operating System Interface.	3	1
	1.4	System Calls, Types of System Calls.	4	1
	Module II - Process and Process Coordination (17 Hours)			

2	**2.1**	Basic Concepts, Process Scheduling-- Scheduling Criteria, Scheduling Algorithms, Multiple Processor Scheduling. , Inter process communication.	5	2
	2.2	Semaphores, Classic Problems of Synchronization, Monitors.	3	2
	2.3	Deadlocks: System Model, Deadlock Characterization, Methods of handling Deadlocks.	4	2
	2.4	Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock.	5	2
3	**Module III - Memory Management (17 Hours)**			
	3.1	Memory Management Strategies - Contiguous memory allocation, swapping	3	3
	3.2	Paging- Concept and purpose of paging ,Logical to physical address translation using page tables, Advantages and disadvantages of paging, Example problems on address translation.	5	3
	3.3	Segmentation – Concepts and purpose of segmentation, Address translation in segmentation.	5	3
	3.4	Virtual Memory Management- Demand paging, Page Replacement.	4	3
4	**Practical Session**			
	Program to illustrate Linux Commands			
	4.1	1. Implement Linux directory commands 2. Implement Linux File commands 3. Implement Linux User Commands 4. Implement Linux filter commands	10	4
	Program to implement shell programming			

	4.2	1. Write a shell script to display echo command with options. 2. write a shell script to perform arithmetic operations (+,-,*,/,++,--) 3. Write a shell script to find the largest among the 3 given numbers 4. Write a program to determine whether the input character is capital or small letter, digits or special symbol. 5. Write a shell program to reverse the digits of five digit integer 6. Write a shell program to find factorial of given number. 7. write a program to find out sum of prime numbers up to a limit 8. Write shell program to create arithmetic calculator using switch. 9. write a Linux program to print Fibonacci series up to the limit using function 10. Write a shell program to concatenate two strings and find the length of the resultant string 11. Write a shell program to add, subtract and multiply the 2 given numbers passed as command line arguments.	20	4
5	**(Teacher Specific Content)**			

<table>
<tr><td>Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)
● Use of ICT tools in conjunction with traditional classroom teaching methods
● Interactive sessions
● Class discussions</td></tr>
<tr><td rowspan="3">Assessment Types</td><td>MODE OF ASSESSMENT
Mode of Assessment</td></tr>
<tr><td>A. Continuous Comprehensive Assessment (CCA)
CCA for Practical (30 Marks)
1. Lab Assessment
2. Observation of Practical Skills
3. On time submission of record
4. Viva</td></tr>
<tr><td>B. End Semester Evaluation (ESE)
ESE for Practical : Practical Examination(70 Marks,3 hrs)
Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks)
Part B : Practical Questions (2 Questions) – (2*15=30 Marks)
Viva : 10 Marks
Record : 10 Marks</td></tr>
</table>

REFERENCES

1. Silberschatz, A., Galvin, P. B., & Gagne, G. (2018). Operating system concepts (10th ed.). Wiley.
2. Tanenbaum, A. S., & Bos, H. (2015). Modern operating systems (4th ed.). Pearson.

SUGGESTED READINGS

1. William E.Shotts,Jr.(2019)-The Linux Command Line: A Complete Introduction (2nd ed.).
2. Dhamdhere, D. M. (2017). Operating systems: A concept-based approach (3rd ed.). McGraw-Hill Education.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences/ Software Development and System Administration				
Programme	B.Voc.(Honours) Software Development and System Administration				
Course Name	.NET & SQL Server – Desktop Application Development				
Type of Course	MPC				
Course Code	MG4MPCSDS200				
Course Level	200				
Course Summary	This course teaches students how to design, develop, and deploy desktop-based applications using Microsoft.NET Framework and SQL Server. It covers UI design, data binding, event-driven programming, form validation, ADO.NET, and database operations.				
Semester	4	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre- requisites, if any	NIL				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the fundamentals of .NET framework and C# programming.	U	1,2,3
2	Learn to build Windows applications using .NET.	A	1,2,3
3	Gain proficiency in SQL for data storage and retrieval.	A	1,2,3,9,10

4	Develop real-world applications integrating .NET and SQL databases	C	1,2,3,9,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/P O	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**2**	**2**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
CO 2	**2**	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
CO 3	**3**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**1**	**3**
CO 4	**3**	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**2**	**3**

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COURSE CONTENT

Content for Classroom transaction (Units)

Module	**Units**	**Course description**	**Hrs**	**CO No.**
Upon completion of this course students will able to understand:				
	Module I: Introduction to .NET Framework and C# (15 Hours)			
1	**1.1**	Overview of .NET framework and CLR	3	1
	1.2	Introduction to Visual Studio IDE	3	1
	1.3	Basics of C#: Data types, variables, operators	3	1
	1.4	Control statements: if, switch, Loops	3	1
	1.5	Classes and Objects in C#	3	1
	Module II: Windows Forms and Database Connectivity (15 Hours)			

2	**2.1**	Introduction to Windows Forms	3	2
	2.2	Controls: TextBox, Label, Button, ListBox, etc.	3	2
	2.3	Event-driven programming	3	2
	2.4	ADO.NET: Concepts and architecture	3	2
	2.5	Connecting to SQL databases using ADO.NET	3	2
3	**Module III: SQL Fundamentals and Advanced Queries (15 Hours)**			
	3.1	Introduction to RDBMS and SQL	3	3
	3.2	SQL Commands: DDL, DML, DCL	4	3
	3.3	Creating and managing tables	4	3
	3.4	SELECT queries with WHERE, ORDER BY, GROUP BY	4	3
4	**Practical Module: Application Development using .NET and SQL (30 Hour)**			
	4.1	• Set up a Windows Forms project in .NET • Connect to a SQL database using ADO.NET • Create user interfaces for data input • Implement database operations (Insert, Update, Delete, Search) • Develop a mini-project (e.g., Student Management System, Inventory App)	30	4
5	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment**
	A. Continuous Comprehensive Assessment (CCA) **CCA for Practical (30 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva
	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination(70 Marks,3 hrs)** Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES

1. Troelsen, A., & Japikse, P. (2020). Pro C# 8 with .NET Core 3. Apress.
2. Beaulieu, A. (2009). Learning SQL (2nd ed.). O’Reilly Media.

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1. Liberty, J. (2008). Programming C# (4th ed.). O’Reilly Media.

	Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**				
Programme	**B.Voc.(Honours) Software Development and System Administration**				
Course Name	**MIS Deployment in IT**				
Type of Course	**SEC**				
Course Code	**MG4SECSDS200**				
Course Level	**200**				
Course Summary	This course introduces the fundamentals and applications of Management Information Systems (MIS) in an IT context. It provides students with a strong conceptual foundation of MIS and its role in decision-making and strategic management. The course also highlights real-world applications of MIS in business and public administration, including ethical, security, and global considerations.				
Semester	4	**Credits**		3	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	0	0	45
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand and describe the components and functions of MIS, Identify various types of systems and their roles in business processes	U	1,2,3,10
2	Analyze the role of MIS in decision support and functional areas, Apply MIS concepts to real-world business scenarios through case studies	An	1,2,3,5,6,9,10

3	Evaluate strategic and ethical implications of MIS in organizations; Understand how global and modern IT trends impact MIS.	E	1,2,3,6,7,8,10
Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**2**	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 2	**3**	**3**	**2**	**0**	**2**	**2**	**0**	**0**	**2**	**2**
CO 3	**3**	**2**	**2**	**0**	**0**	**3**	**2**	**3**	**0**	**2**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	**MODULE I – MIS Concepts and Architecture (15 Hours)**			
	1.1	Introduction to MIS: Definition, Importance, Characteristics, Components of MIS: Hardware, Software, Data, Procedures, People, Types of Information Systems: TPS, MIS, DSS, ESS	8	1
	1.2	System Architecture, Data and Information, Role of MIS in Business and Management, Classification of MIS by Functions and Levels	7	1
2	**MODULE II - Decision Support Systems and MIS Applications (15 Hours)**			
	2.1	Decision Making and MIS: Types of Decisions, Decision Support Systems (DSS) and Executive Information Systems (EIS), Business Intelligence, Data Warehousing, Data Mining	8	2

	2.2	Functional Information Systems: Marketing, Finance, HR, Manufacturing, Case Studies of MIS Applications in Real-world Organizations	7	2
3	**MODULE III - Information Systems Strategy and Emerging Trends (15 Hours)**			
	3.1	Strategic Role of Information Systems in Business, Competitive Advantage with IT (Porter's Models, ERP, CRM, SCM), E-Governance and MIS in Public Sector (with focus on Digital India and Digital Kerala)	8	3
	3.2	Security, Privacy, and Ethical Issues in MIS,Global MIS and Emerging Trends (Cloud Computing, IoT, AI in MIS)	7	3
4	**(Teachers specific content)**			

<table>
<tr><td>Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)
Use of ICT tools in conjunction with traditional classroom teaching methods
Interactive sessions
Class discussions</td></tr>
<tr><td rowspan="2">Assessment Types</td><td>MODE OF ASSESSMENT
Mode of Assessment</td></tr>
<tr><td>A. Continuous Comprehensive Assessment (CCA)
CCA for Theory(25 Marks)
Written Test
Assignment
Seminar</td></tr>
</table>

	B. End Semester Evaluation (ESE) – Non-MCQ **ESE for Theory : Written Test(50 Marks,1.5 hrs)** Part A : Very Short Answer Questions (10 out of 12 Questions) - (10*2=20 Marks) Part B : Short Answer Questions (4 out of 6 Questions) - (4*5=20Marks) Part C : Essay Questions (1 out of 3 Questions) - (1*10=10 Marks)

REFERENCES

1. Laudon, K. C., & Laudon, J. P. (2021). Management Information Systems: Managing the Digital Firm (16th ed.). Pearson.
2. O'Brien, J. A., & Marakas, G. (2011). Management information systems (10th ed.). McGraw-Hill

SUGGESTED READINGS

1. Goyal, D. P. (2016). Information systems for managers *(3rd ed.).* Macmillan Publishers.
2. Davis, G. B., & Olson, M. H. (1985). Management information systems: A managerial perspective (2nd ed.). McGraw-Hill.

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<table>
<tr><td>Faculty/ Discipline</td><td colspan="5">Technology and Applied Sciences/ Software Development and System Administration</td></tr>
<tr><td>Programme</td><td colspan="5">B.Voc.(Honours) Software Development and System Administration</td></tr>
<tr><td>Course Name</td><td colspan="5">Social Impact of Technology</td></tr>
<tr><td>Type of Course</td><td colspan="5">VAC</td></tr>
<tr><td>Course Code</td><td colspan="5">MG4VACSDS200</td></tr>
<tr><td>Course Level</td><td colspan="5">200</td></tr>
<tr><td>Course Summary</td><td colspan="5">This course investigates the intersection between technology and society, critically examining the influence of technological advancements on employment, social structures, ethics, economy, and the environment. The course encourages students to think analytically about real-world implications of technology, fostering responsible innovation and ethical awareness.</td></tr>
<tr><td>Semester</td><td>4</td><td colspan="2">Credits</td><td>3</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course Details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>45</td></tr>
<tr><td>Pre-requisites, if any</td><td colspan="5">NIL</td></tr>
</table>

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Analyze how technology influences societal values and cultural practices. Identify ethical challenges in the deployment of digital technologies.	U	1,2,3,6,7,8,10
2	Evaluate the effects of automation and AI on employment trends. Pro bridging the technology-related skills gap.	U	1,2,3,6,7,8,10

3	Assess the environmental consequences of technological products and services. Recommend sustainable practices in technology design and usage.	A	1,2,3,6,7,8,10
Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**3**	**2**	**2**	**0**	**0**	**3**	**3**	**3**	**0**	**2**
CO 2	**3**	**3**	**2**	**0**	**0**	**2**	**2**	**2**	**0**	**3**
CO 3	**2**	**2**	**2**	**0**	**0**	**3**	**3**	**2**	**0**	**2**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I - Technology, Culture, and Ethics (14 Hours)**			
	1.1	Historical evolution of technology and its impact on society Role of technology in shaping culture and Communication	7	1
	1.2	Digital divide and access disparities, Technology and social change, Surveillance, privacy, and ethics	7	
2	**Module II - Digital Economy and Future Work (15 Hours)**			
	2.1	Automation and the future of work, Artificial intelligence and labor displacement, Gig economy and platform work	7	2
	2.2	Technology-induced skills gap, Digital entrepreneurship and job creation	8	2
	Module III - Sustainable Technology and Green Computing (16 Hours)			

3	**3.1**	E-waste and sustainable computing, Green IT and carbon footprint, Ethical sourcing of materials	8	3
	3.2	Technology's role in climate monitoring and disaster response, Smart cities and sustainable development	8	3
4	**(Teacher specific contents)**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • se of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment**
	A. Continuous Comprehensive Assessment (CCA) **CCA for Theory(25 Marks)** 1. Written Test 2. Assignment 3. Seminar
	B. End Semester Evaluation (ESE) – Non-MCQ **ESE for Theory : Written Test(50 Marks,1.5 hrs)** Part A : Very Short Answer Questions(10 out of 12 Questions) – (10*2=20 Marks) Part B : Short Answer Questions(4 out of 6 Questions) - (4*5=20 Marks) Part C : Essay Questions(1 out of 3 Questions) - (1*10=10 Marks)

REFERENCES

1. Crawford, K. (2021*).* Atlas of AI: Power, politics, and the planetary costs of artificial intelligence. Yale University Press.
2. Brynjolfsson, E., & McAfee, A. (2011). Race against the machine: How the digital revolution is accelerating innovation, driving productivity, and irreversibly transforming employment and the

economy. Digital Frontier Press.
3. Winston, B. (1998). Media technology and society: A history from the telegraph to the internet. Psychology Press.

SUGGESTED READINGS

1. Brynjolfsson, E., & McAfee, A. (2011). Race against the machine: How the digital revolution is accelerating innovation, driving productivity, and irreversibly transforming employment and the economy. Digital Frontier Press.

	Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**Summer Internship**
Type of Course	**INT**
Course Code	**MG4INTSDS200**
Course Summary	The internship is designed to provide students with real-world exposure and hands-on experience in professional environments aligned with their skill domain and major area of study. It acts as a vital link between academic learning and industry application, allowing students to apply theoretical concepts to practical situations. Through active engagement in industry, research institutions, or academic labs, students gain insights into organizational operations, workplace practices, and professional expectations. The internship also supports the development of key professional competencies such as communication, teamwork, time management, and ethical responsibility. Additionally, it encourages critical thinking, reflection, and self-assessment, helping students identify personal strengths and explore potential career pathways. Students shall undergo the internship in a Firm, Industry, or Organization, or engage in Training in Labs with faculty and researchers, or other Higher Education or Research Institutions, ensuring alignment with their area of academic specialization.

Semester	**4**	**Duration**	**60 hours**	**Credits**	**2**

COURSE OUTCOMES (CO)

CO No:	**Expected Course Outcome**	**Learning Domains**	**PO No:**
1	Demonstrate practical understanding of operational aspects in Their domain by engaging in real-world industry settings.	Ap	1,3,6,10

9

2	Apply academic knowledge and skills to identify and solve industry-relevant problems.	A	1,2,3,10
3	Exhibit professional competencies including effective communication, teamwork, time management, and ethical Responsibility.	S	4,5,8,9
4	Develop an understanding of workplace practices, expectations, and challenges.	U	1,6,10
5	Reflect critically on their internship experience to identify Personal strengths, growth areas, and career aspirations.	E	1,6,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**3**	**0**	**3**	**0**	**0**	**2**	**0**	**0**	**0**	**2**
CO 2	**3**	**2**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 3	**0**	**0**	**0**	**3**	**2**	**0**	**0**	**2**	**2**	**0**
CO 4	**3**	**0**	**0**	**0**	**0**	**2**	**0**	**0**	**0**	**2**
CO 5	**3**	**0**	**0**	**0**	**0**	**2**	**0**	**0**	**0**	**2**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

Assessment Types	**MODE OF ASSESSMENT**		
	A	**Continuous Comprehensive Assessment (CCA)**	
		Components	**Marks**
		Feedback from the hosting organization	5
		Internal Supervisor feedback	10
		Total	**15**
	B	**End Semester Evaluation (ESE)**	
		Components	**Marks**
		Presentation	10
		Report	10
		Viva Voce	15
		Total	**35**

SEMESTER V

Semester: 5

Course Code	Title of the Course	Type of the Course	Credit	Hours/ Week	Hour Distribution /week		
					L	P	O
MG5SDCSDS300	Cloud Computing & DevOps	SDC	4	4	4	0	0
MG5SDCSDS301	React.js	SDC	4	5	3	2	0
MG5SDESDS300	Node.js	SDE (Elective Papers – Select one)	4	5	3	2	0
MG5SDESDS301	ASP .NET Model View Controller Application						
MG5SDESDS302	Enterprise Database Administration with Advanced SQL Server						
MG5SECSDS300	Computer Hardware Maintenance & Troubleshooting	SEC	3	4	2	2	0
MG5MPCSDS300	Programming with Python	MPC	4	4	4	0	0
MG5VACSDS300	Cyber Laws & Online Security	VAC	3	3	3	0	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

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Faculty/ Discipline	**Technology and Applied Sciences**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**Cloud Computing & DevOps**
Type of Course	**SDC**
Course Code	**MG5SDCSDS300**
Course Level	**300**
Course Summary	This course teaches students about Cloud Computing and DevOps practices, essential for modern IT infrastructure. It covers cloud models, deployment models, and core services from major providers. It also covers CI/CD tools, automation, monitoring, and infrastructure as code, ensuring efficient design, implementation, and management of cloud-based applications.

Semester	5	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		4	0	0	60
Pre-requisites, if Any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Foundations of Cloud Computing	U	1,2,3,9,10
2	Cloud Service Platforms and Architectures	An	1,2,3,5,9,10
3	DevOps Fundamentals and Lifecycle Management	U	1,2,3,5,8,9,10
4	DevOps Tools and Implementation Techniques	S	1,2,3,5,8,9,10

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	3	2	0	0	0	0	0	1	2
CO 2	2	3	2	0	1	0	0	0	2	2
CO 3	2	3	2	0	1	0	0	1	3	3
CO 4	2	3	2	0	1	0	0	1	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I - Introduction to Cloud Computing (15 Hours)**			
	1.1	Historical developments, characteristics of cloud computing	4	1
	1.2	Pros and Cons of cloud computing	3	1
	1.3	Building cloud computing environments, Types of cloud,	4	1
	1.4	Parallel vs. Distributed Computing, Virtualization and cloud computing.	4	1
2	**Module II - Cloud platforms in industry (15 Hours)**			
	2.1	Amazon Web Services- compute services, storage services, communication services	6	2
	2.2	Google AppEngine - architecture and core concepts, application life cycle	5	2
	2.3	Microsoft Azure – Azure core concepts.	4	2
	Module III - DevOps Fundamentals and Lifecycle (15 Hours)			

3	3.1	**Introduction to DevOps-** Definition, History, and Evolution of DevOps, Differences between Traditional SDLC, Agile, and DevOps, Business and technical benefits of DevOps	5	3
	3.2	**DevOps Culture and Principles-** Collaboration, Feedback Loops, and Automation,DevOps vs Agile vs ITIL vs SRE, The CALMS model (Culture, Automation, Lean, Measurement, Sharing)	5	3
	3.3	**DevOps Lifecycle-** Overview of Phases: Plan, Develop, Build, Test, Release, Deploy, Operate, Monitor Key DevOps Metrics: Deployment frequency, lead time, MTTR, change failure rate Introduction to DevOps pipeline and workflow orchestration	5	3
4	**Module IV - DevOps Tools and Practices (15 Hours)**			
	4.1	**Version Control Systems-** Introduction to Git, Git workflows (feature branch, fork & clone, pull request)	5	4
	4.2	**Continuous Integration / Continuous Deployment (CI/CD)**-CI/CD principles and pipeline structure, Tools: Jenkins (basics), GitHub Actions (overview)	5	4
	4.3	**Containerization and Virtualization-** Docker basics: images, containers, Dockerfile, Overview of orchestration tools: Kubernetes (brief introduction)	5	4
5	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions	
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment**	
	A. Continuous Comprehensive Assessment **(CCA) CCA for Theory(30 Marks)** 1. Written Test 2. Seminars 3. Assignments	
	B. End Semester Evaluation (ESE) – Non-MCQ **ESE for Theory : Written Test(70 Marks , 2 hrs)** Part A: Very Short Answer Questions (10 out of 12 Questions)- (10*2=20 Marks) Part B: Short Answer Questions (4 out of 6 Questions) - (4*5=20 Marks) Part C:Essay Questions(2 out of 4 Questions) - (2*15=30 Marks)	

REFERENCES

1. RajkumarBuyya, Christian Vecchiola, S ThamaraiSelvi- Mastering Cloud Computing, Tata McGraw Hill Publications.
2. A Srinivasan& J. Suresh " Cloud Computing : A Practical Approach for learning and Implementation " , First edition ,Pearson
3. Bass, L., Weber, I., & Zhu, L. (2015). DevOps: A software architect's perspective (1st ed.). Addison-Wesley Professional.

SUGGESTED READINGS

1. Kumar Saurabha, "Cloud Computing " Wiley Publication Krutz ,Vines "Cloud Security". Wiley Publication.
2. Kim, G., Behr, K., & Spafford, G. (2013). The Phoenix Project: A novel about IT, DevOps, and helping your business win. IT Revolution Pres

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<table>
<tr><td>Faculty/ Discipline</td><td colspan="5">Technology and Applied Sciences/ Software Development and System Administration</td></tr>
<tr><td>Programme</td><td colspan="5">B.Voc.(Honours) Software Development and System Administration</td></tr>
<tr><td>Course Name</td><td colspan="5">React.js</td></tr>
<tr><td>Type of Course</td><td colspan="5">SDC</td></tr>
<tr><td>Course Code</td><td colspan="5">MG5SDCSDS301</td></tr>
<tr><td>Course Level</td><td colspan="5">300</td></tr>
<tr><td>Course Summary</td><td colspan="5">This course teaches students React.js fundamentals for building dynamic user interfaces, particularly for single-page applications, through practical projects and hands-on exposure.</td></tr>
<tr><td>Semester</td><td>5</td><td colspan="2">Credits</td><td>4</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course Details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>75</td></tr>
<tr><td>Pre-requisites, if any</td><td colspan="5">NIL</td></tr>
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COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Describe the core concepts and environment of React.js	U	1,2,3,10
2	Use components, props, and JSX to create user interfaces	A	1,2,3,8,9,10
3	Manage state and events in React applications using hooks	A	1,2,3,8,9,10
4	Implement routing and fetch API data in React apps	A	1,2,3,8,9,10

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	2	1	0	0	0	0	0	0	3
CO 2	2	3	2	0	0	0	0	1	1	3
CO 3	3	3	2	0	0	0	0	1	1	3
CO 4	3	3	2	0	0	0	0	1	2	3

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COURSE CONTENT

Content for Classroom Transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I – Getting Started with React (15 Hours)**			
	1.1	What is React.js and why use it?	3	1
	1.2	Installing Node.js, npm, and Create React App (CRA)	3	1
	1.3	Introduction to JSX syntax	3	1
	1.4	Functional components and rendering	3	1
	1.5	Props and PropTypes	3	1
2	**Module II – State, Events, and Hooks (15 Hours)**			
	2.1	Introduction to state and useState hook	2	2
	2.2	Handling user events (onClick, onChange)	3	2

	2.3	Conditional rendering	2	2
	2.4	List rendering using .map() and key prop	2	2
	2.5	Controlled vs uncontrolled components	2	2
	2.6	Basic form handling	2	2
	2.7	Introduction to useEffect	2	2
3	**Module III – Routing and Basic API Handling (15 Hours)**			
	3.1	Introduction to React Router (v6)- BrowserRouter, Routes, Route, Link and useNavigate	3	3
	3.2	Dynamic routing basics	3	3
	3.3	Introduction to API calls using fetch()	3	3
	3.4	Displaying data from JSON API	3	3
	3.5	Basic error handling and loading states	3	3
4	**Module IV – Practical Session (30 Hours)**			
	4.1	1. Install Node.js and Create React App, then display "Hello React" in the browser. 2. Make a separate Header component that takes a title as a prop and displays it. 3. Create a counter with “Increase” and “Decrease” buttons using useState. 4. Add a button to show/hide a text message when clicked. 5. Store a list of products in state and display them using .map() with a key. 6. Make a form with name and email fields using controlled components,	30	4

		and display the submitted data. 7. Use useEffect to create a timer that starts counting seconds when the page loads. 8. Use React Router to create two pages: Home and About, and navigate between them. 9. Get data from https://jsonplaceholder.typicode.com/users and display it in a table. 10. Show "Loading..." while fetching and display "Error loading data" if the fetch fails.		
5	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Use of ICT tools in conjunction with traditional classroom teaching methods ● Interactive sessions ● Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment** **A. Continuous Comprehensive Assessment (CCA)** **CCA for Practical (30 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva

	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination(70 Marks,3 hrs)** Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES

1. Banks, A., & Porcello, E. (2020). Learning React (2nd ed.). O'Reilly Media.
2. Accomazzo, A., Murray, N., Lerner, A., Greathouse, C., & Auerbach, D. (2021). Fullstack React: The complete guide to ReactJS and friends (2nd ed.). Fullstack.io.

SUGGESTED READINGS

1. Stefanov, S. (2016). React: Up and running: Building web applications. O'Reilly Media.
2. Grider, S. (2019). React: Up & running: Building web applications (2nd ed.). O'Reilly Media.

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<tr><td>Programme</td><td colspan="5">B.Voc.(Honours) Software Development and System Administration</td></tr>
<tr><td>Course Name</td><td colspan="5">Node.js</td></tr>
<tr><td>Type of Course</td><td colspan="5">SDE</td></tr>
<tr><td>Course Code</td><td colspan="5">MG5SDESDS300</td></tr>
<tr><td>Course Level</td><td colspan="5">300</td></tr>
<tr><td>Course Summary</td><td colspan="5">This course introduces Node.js for backend web development. Students will learn to build dynamic, database-driven web applications using Node.js, Express.js, and MySQL, with focus on user interaction, data handling, and session management.</td></tr>
<tr><td>Semester</td><td>5</td><td colspan="2">Credits</td><td>4</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course Details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>75</td></tr>
<tr><td>Pre-requisites, if Any</td><td colspan="5">NIL</td></tr>
</table>

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the core concepts of Node.js and Express.js, including its architecture, core modules, routing, middleware, and templating engines, to develop basic server-side applications.	U	1,2,3,10
2	Apply server-side programming techniques to handle form data, establish MySQL connectivity, and perform CRUD operations using Node.js and Express.js.	A	1,2,3,10

3	Design and implement secure authentication systems with session management and role-based access control in a Node.js and Express.js application.	C	1,2,3,8,10
4	Develop and deploy a full-stack web application integrating user authentication, MySQL database operations, and administrative functionality using the Node.js ecosystem	C	1,2,3,8,9,10
Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S),Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**2**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 2	**1**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**3**
CO 3	**1**	**2**	**1**	**0**	**0**	**0**	**0**	**3**	**0**	**2**
CO 4	**1**	**2**	**3**	**0**	**0**	**0**	**0**	**1**	**2**	**2**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I - Fundamentals of Node.js and Express.js (15 Hours)**			
	1.1	Introduction to Node.js: Features, use cases, installation, Node.js architecture: Single-threaded, event-driven model	3	1
	1.2	Creating a basic HTTP server, Core Modules (fs, http, url, path)	3	1
	1.3	npm and package management	2	1

	1.4	Introduction to Express.js, Installing and setting up Express, Middleware, URL routing, static files	4	1
	1.5	Template engines (Handlebars - HBS), Building a form-based application	3	1
2	**Module II - Forms and MySQL Integration (15 Hours)**			
	2.1	Handling form data and HTTP methods	3	2
	2.2	Connecting Node.js with MySQL	2	2
	2.3	Creating tables, managing connections	3	2
	2.4	Executing SQL queries through Node.js	3	2
	2.5	Performing CRUD operations: Insert, View, Edit, Delete	4	2
3	**Module III -Authentication and Session Management (15 Hours)**			
	3.1	User login/logout	3	3
	3.2	Session management using Express-session	3	3
	3.3	Authentication logic: role-based access (admin/user)	5	3
	3.4	Securing routes, Flash messages and error handling	4	3
	Practical (30 Hours)			

4	**4.1**	1. Create a simple Node.js server 2. Use core modules to read/write files and serve basic responses 3. Build a basic Express.js application 4. Create Login and Registration pages using HBS 5. Setup project file structure and render views 6. Insert user data from registration form to MySQL 7. Display user list using dynamic views 8. Implement Edit and Delete functions 9. Implement login and session handling 10. Create protected views based on user roles	30	4
5	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Use of ICT tools in conjunction with traditional classroom teaching methods ● Interactive sessions ● Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment** **A. Continuous Comprehensive Assessment (CCA)** **CCA for Practical (30 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva

	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination(70 Marks,3 hrs)** Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES

1. Cantelon, M., Harter, M., Holowaychuk, T., & Rajlich, N. (2014).Node.js in action (2nd ed.). Manning Publications.
2. Brown, E. (2020).Learning JavaScript: JavaScript Essentials for Modern Application Development (3rd ed.). O'Reilly Media.
3. O'Hara, B. (2019).Web Development with Node and Express: Leveraging the JavaScript Stack (2nd ed.). O'Reilly Media.

SUGGESTED READINGS

1. Pasquali, S., & Faaborg, K. (n.d.). Mastering Node.js (2nd rev. ed.). Packt Publishing.
2. Herron, S. (2019). Node.js web development (4th ed.). Packt Publishing.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**ASP .NET Model View Controller Applications**
Type of Course	**SDE**
Course Code	**MG5SDESDS301**
Course Level	**300**
Course Summary	This course introduces students to building dynamic web applications using the ASP.NET MVC framework. It covers MVC architecture, controller logic, view rendering with HTML Helpers, and database integration using ADO.NET and SQL Server. Learners will gain hands- on experience in developing and validating interactive, data-driven web applications.

Semester	5	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the MVC architecture and how the design pattern is implemented in ASP.NET MVC, including the roles of Model, View, and Controller.	U	1,2,3
2	Apply HTML Helpers, Controller methods to develop interactive, and data- bound user interfaces using MVC framework.	A	1,2,3

3	Design and implement SQL Server databases using ADO.NET and perform data operations using SQL queries and stored procedures.	An	1,2,3,10
4	Develop complete ASP.NET MVC web applications that integrate database operations, strong validation, and dynamic data handling using practical knowledge.	A	1,2,3,10
Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**3**	**2**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
CO 2	**3**	**2**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
CO 3	**2**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 4	**2**	**3**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**3**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I - Fundamentals of MVC Architecture and ASP.NET MVC Framework (15 Hours)**			
	1.1	Introduction to MVC Architecture	3	1
	1.2	Understand the MVC design pattern and its application in ASP.NET MVC	4	1
	1.3	Understanding Model, View, and Controller concepts	4	1
	1.4	Key benefits of ASP.NET MVC and advantages of MVC-based web applications	4	1
2	**Module II - Building Interactive Web Interfaces with HTML Helpers and Controllers in ASP.NET MVC (15 Hours)**			

	2.1	HTML Helper Methods: Rendering HTML form elements and dropdown lists	2	2
	2.2	Binding HTML Helpers to Model, using "For" methods with typed models	2	2
	2.3	Creating Views with HTML Helpers	2	2
	2.4	Exploring Controllers and the Controller Base class	3	2
	2.5	Passing data from Controller to View, comparing ViewData, ViewBag, and TempData	3	2
	2.6	Types of Action Methods, Action Method Parameters, and Model Binders	3	2
3	**Module III - Database Integration and Data Handling in ASP.NET MVC Applications (15 Hours)**			
	3.1	Introduction to ADO.NET and creating database tables with relationships	2	3
	3.2	SQL Fundamentals – Insert, Update, Delete, Select commands	3	3
	3.3	Working with Stored Procedures	3	3
	3.4	Creating strongly-typed views and understanding URL patterns and action methods	3	3
	3.5	Using HTML Helpers, handling form post-backs, and implementing data validation	4	3
4	**(Practical Session)**			
	4.1	1. Create a basic MVC application 2. Demonstrate data passing from controller to view 3. Create forms using HTML helpers 4. Implement form submission and post-back 5. Connect MVC app with SQL Server 6. Perform Insert, Update, Delete operations 7. Use stored procedures in MVC 8. Validate input data in form submission	30	4
5	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment**
	A. Continuous Comprehensive Assessment (CCA) **CCA for Practical (30 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva
	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination(70 Marks,3 hrs)** Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES

1. Esposito, D. (2014). Programming ASP.NET MVC 5. Microsoft Press.
2. Freeman, A. (2018). Pro ASP.NET Core MVC 2 (7th ed.). Apress.

SUGGESTED READINGS

1. Rahien, A., & Hanselman, S. (2013). Professional ASP.NET MVC 5.
2. Penrod, J., Johnson, A., & Pedersen, J. (2017). Professional ASP.NET MVC 5 (1st ed.). Wrox.

	Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**				
Programme	**B.Voc.(Honours) Software Development and System Administration**				
Course Name	**Enterprise Database Administration with SQL Server**				
Type of Course	**SDE**				
Course Code	**MG5SDESDS302**				
Course Level	**300**				
Course Summary	The Enterprise Database Administration with SQL Server course provides foundational knowledge and hands-on skills for managing, securing, and maintaining SQL Server databases. It covers user authentication, backup and recovery, performance monitoring, and data import/export. Learners gain practical experience in configuring SQL Server environments for reliability, security, and efficiency.				
Semester	5	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand user authentication, authorization, and role-based access control in SQL Server.	U	1,2,3
2	Configure and apply backup, recovery, auditing, and SQL Server Agent features for securing and maintaining database environments.	A	1,2,10

3	Monitor, troubleshoot, optimize SQL Server performance, and manage data import/export processes effectively.	An	1, 2, 3,10
4	Implement practical administrative tasks including security setup, automation, backup/restore, monitoring, and deployment of databases.	A	1, 2, 3,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	1	0	0	0	0	0	0	0
CO 2	3	2	0	0	0	0	0	0	0	1
CO 3	2	3	1	0	0	0	0	0	0	2
CO 4	2	3	2	0	0	0	0	0	0	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I - Securing SQL Server Access and Roles (9 Hours)**			
	1.1	Authenticate and authorize users in SQL Server	3	1
	1.2	Assign server-level and database-level roles	3	1
	1.3	Authorize users to access database resources securely	3	1
2	**Module II - Data Protection, Backup, and Security Agent Configuration (16 Hours)**			
	2.1	Protect sensitive data using encryption and auditing	3	2
	2.2	Describe SQL Server recovery models and backup strategies	2	2

	2.3	Perform full and differential SQL Server backups	3	2
	2.4	Restore SQL Server databases using different recovery options	3	2
	2.5	Configure SQL Server Agent security and credentials	2	2
	2.6	Manage alerts and notifications using Database Mail, Operators, and Alerts	3	2
3	**Module III - Monitoring, Performance Tuning, and Data Operations (20 Hours)**			
	3.1	Trace access and activity on SQL Server	2	3
	3.2	Monitor SQL Server infrastructure health and performance	2	3
	3.3	Capture and manage performance data using built-in tools	3	3
	3.4	Analyze collected performance metrics for optimization	2	3
	3.5	Troubleshoot SQL Server performance and availability issues	2	3
	3.6	Import and export data using wizards and command-line tools (bcp, BULK INSERT)	4	3
	3.7	Deploy and upgrade Data-Tier Applications (DACPAC)	5	3
4	**(Practical Session)**			
	4.1	1. Configure server and database roles 2. Set up data encryption and auditing 3. Perform backup and restore operations 4. Configure SQL Server Agent and Database Mail 5. Monitor SQL Server performance 6. Import/export data using bcp and BULK INSERT7. Deploy a Data-Tier Application	30	4
5	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions

<table>
<tr><td rowspan="3">Assessment Types</td><td>MODE OF ASSESSMENT
Mode of Assessment</td></tr>
<tr><td>A. Continuous Comprehensive Assessment (CCA)
CCA for Practical (30 Marks)
1. Lab Assessment
2. Observation of Practical Skills
3. Viva</td></tr>
<tr><td>B. End Semester Evaluation (ESE)
ESE for Practical : Practical Examination(70 Marks,3 hrs)
Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks)
Part B : Practical Questions(2 Questions) – (2*15=30 Marks)
Viva : 10 Marks
Record : 10 Marks</td></tr>
</table>

REFERENCES

1. Lahdenmaki, R., & Leach, M. (2005). Relational database index design and the optimizers. Wiley.
2. Jorgensen, K. (2017). Microsoft SQL Server 2017 administration inside out. Microsoft Press.
3. Petkovic, D. (2019). Microsoft SQL Server 2019: A beginner's guide (7th ed.). McGraw-Hill Education.

SUGGESTED READINGS

1. Jorgensen, K. (2018). *Microsoft SQL Server 2017 administration inside out*. Microsoft Press.
2. Jorgensen, K., Machanic, A., & Brewer, G. (2020). *SQL Server 2019 administrator's guide* (2nd ed.). Packt Publishing.
3. Petkovic, D. (2020). *Microsoft SQL Server 2019: A beginner's guide* (7th ed.). McGraw-Hill Education.

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Faculty/ Discipline	**Technology and Applied sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**Computer Hardware Maintenance & Troubleshooting**
Type of Course	**SEC**
Course Code	**MG5SECSDS300**
Course Level	**300**
Course Summary	This course introduces students to the fundamentals of computer hardware, its common issues, and hands-on troubleshooting techniques. With a balance of theoretical knowledge and practical sessions, students will gain the confidence to assemble, diagnose, maintain, and repair computer systems. This will help them support IT infrastructure in workplaces or pursue careers in technical support, system administration, or field service.

Semester	5	**Credits**		3	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		2	1	0	60
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the basic structure, components, and functioning of computer hardware.	U	1,2,3,6,8,9,10
2	Identify and troubleshoot common computer hardware problems.	A	1,2,3,5,6,8,9,10
3	Perform hands-on maintenance and repair of basic desktop hardware systems.	S	1,2,3,5,6,8,9,10

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	2	2	0	0	1	0	1	1	3
CO 2	3	3	2	0	1	1	0	1	1	3
CO 3	3	3	2	0	1	1	0	1	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I - Basics of Computer Hardware (15 Hours)**			
	1.1	Introduction to computer hardware: CPU, motherboard, RAM, SMPS, hard disk, SSD, optical drives	3	1
	1.2	Input and Output devices: Keyboard, mouse, monitor, printer, scanner	2	1
	1.3	Types of computers: Desktop, laptop, server	2	1
	1.4	Ports and connectors: USB, HDMI, VGA, audio, LAN	3	1
	1.5	BIOS/UEFI and POST process	2	1
	1.6	Safety precautions in handling hardware	3	1
2	**Module II - Hardware Troubleshooting Techniques (15 Hours)**			
	2.1	Identifying hardware issues: No display, beep codes, overheating, power failure	3	2
	2.2	RAM, SMPS, motherboard, and hard disk fault diagnosis	2	2
	2.3	Error codes and BIOS messages	2	2

	2.4	Preventive maintenance: Dust cleaning, cable check, thermal paste	3	2
	2.5	Introduction to troubleshooting tools: Multimeter, POST card, diagnostic software	3	2
	2.6	Replacing or upgrading RAM, HDD/SSD, power supply	2	2
3	**Hands-on Session in PC Assembly, Maintenance and Troubleshooting (30 Hours)**			
	3.1	Assembling and disassembling a desktop computer Connecting and testing components Diagnosing hardware faults through trial and error Replacing faulty parts BIOS settings and basic firmware update Using software tools to check hardware health (e.g., CPU-Z, HWMonitor)Creating bootable devices and installing OS for testing	30	3
4	**Teacher Specific Contents**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
	MODE OF ASSESSMENT **Mode of Assessment**

Assessment Types	**A. Continuous Comprehensive Assessment (CCA)** **CCA for Practical(25 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva
	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination (50 Marks, 1.5 hrs)** Part A : Short Answer Questions(5 out of 7 Questions)-(5*4=20 Marks) Part B : Practical Question (1 Question) – (1*20=20 Marks) Record : 10 Marks

REFERENCES

1. Mueller, S. (2015). Upgrading and Repairing PCs (22nd ed.). Que Publishing.
2. Bigelow, S. P. (2011). Troubleshooting, Maintaining and Repairing PCs (6th ed.). McGraw-Hill Education.

SUGGESTED READINGS

1. Meyers, M. (2022). CompTIA A+ Certification All-in-One Exam Guide (Exams 220-1101 & 220-1102) (11th ed.). McGraw-Hill Education.
2. Andrews, J. A. (2020). A+ Guide to IT Technical Support (10th ed.). Cengage Learning

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<table>
<tr><td>Faculty/ Discipline</td><td colspan="5">Technology and Applied Sciences/ Software Development and System Administration</td></tr>
<tr><td>Programme</td><td colspan="5">B.Voc.(Honours) Software Development and System Administration</td></tr>
<tr><td>Course Name</td><td colspan="5">Programming with Python</td></tr>
<tr><td>Type of Course</td><td colspan="5">MPC</td></tr>
<tr><td>Course Code</td><td colspan="5">MG5MPCSDS300</td></tr>
<tr><td>Course Level</td><td colspan="5">300</td></tr>
<tr><td>Course Summary</td><td colspan="5">Understand the basic concepts, syntax, and structures of Python. Apply control structures, functions, and modules to solve programming tasks. Work with Python data structures and perform file operations. Create and test practical Python applications through hands-on programming.</td></tr>
<tr><td>Semester</td><td>5</td><td colspan="2">Credits</td><td>4</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course Details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>4</td><td>0</td><td>0</td><td>60</td></tr>
<tr><td>Pre- requisites, if any</td><td colspan="5">NIL</td></tr>
</table>

COURSE OUCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the history, features, and environment setup of Python, Apply Python syntax and use basic data types and operator, Implement simple interactive programs using input and output operations	U	1,2,3,10

2	Use conditional and looping constructs to control program flow Develop reusable code using user-defined functions Utilize recursion and lambda expressions for efficient coding	A	1,2,3,4,10
3	Use and manipulate various built-in data structures in Python, Apply iteration and comprehension techniques, appropriate data structure usage for specific scenarios	U	1,2,3,4,10
4	Perform file operations for data persistence Handle runtime errors using exception handling constructs Import and use modules to modularize programs.	An	1,2,3,4,10
****Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)***			

CO-PO ARTICULATION MATRIX

CO/P O	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**2**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 2	**2**	**3**	**1**	**1**	**0**	**0**	**0**	**0**	**0**	**2**
CO 3	**2**	**3**	**1**	**1**	**0**	**0**	**0**	**0**	**0**	**3**
CO 4	**2**	**3**	**1**	**1**	**0**	**0**	**0**	**0**	**0**	**3**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	**Units**	**Course description**	**Hrs**	**CO No.**
	Module I – Introduction to Python Programming (16 Hours)			
1	**1.1**	Python Overview: Features, Advantages, Versions, Installation and Environment Setup, Basic Syntax, Identifiers, Indentation, Comments	8	1

	1.2	Variables and Data Types: Numbers, Strings, Booleans, None Arithmetic, Relational, Logical, Assignment, Type Conversion, Input/Output Functions	8	1
2	**Module II - Python Control Structures and Functions (16 Hours)**			
	2.1	Control Flow: if, if-else, if-elif-else Loops: while, for, nested loops, range, Loop Control Statements: break, continue, pass	8	2
	2.2	Functions: Definition, Arguments, Return Statement, Scope and Lifetime of Variable, Recursion	8	2
3	**Module III - Python Data Structures and Operations (18 Hours)**			
	3.1	Strings: Operations, Slicing, Built-in Methods, Lists: Creation, Indexing, Slicing, Methods, Tuples: Properties, Indexing, Operations Sets: Creation, Operations, Built-in Methods.	8	3
	3.2	Dictionaries: Key-Value Pairs, Methods, Iteration over data structures, List/Set/Dictionary Comprehension.	10	3
4	**Module IV - Python Modules and Packages (10 Hours)**			
	4.1	Creating and importing user-defined modules, Packages and _init_.py.	**7**	**4**
	4.2	Python Standard Library overview.	**3**	**4**
5	**(Teacher Specific Content)**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Use of ICT tools in conjunction with traditional classroom teaching methods ● Interactive sessions ● Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment**
	A. Continuous Comprehensive Assessment (CCA) **CCA for Practical (30 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva
	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination(70 Marks,3 hrs)** Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES

1. Downey, A. B. (2015). Think Python: How to think like a computer scientist (2nd ed.). O’Reilly Media.
2. Zelle, J. (2017). Python programming: An introduction to computer science. Franklin, Beedle & Associates.
3. Summerfield, M. (2009). Programming in Python 3. Pearson Education.

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1. Lutz, M. (2013). Learning Python (5th ed.). O’Reilly Media.
2. Matthes, E. (2019). Python crash course (2nd ed.). No Starch Press.
3. Sweigart, A. (2019). Automate the boring stuff with Python (2nd ed.). No Starch Press.

Mahatma Gandhi University Kottayam

Faculty/Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**Cyber Laws & Online Security**
Type of Course	**VAC**
Course Code	**MG5VACSDS300**
Course Level	**300**
Course Summary	This comprehensive course on Cyber Laws and Security is designed to provide participants with a thorough understanding of cyber laws, including the IT Act, data protection, and regulations related to cybercrimes, cyberbullying, and harassment, along with internet security practices. It also provides a foundational understanding of cryptography, cyber forensics, and ethical hacking principles to enhance knowledge in securing digital information and systems.

Semester	5	**Credits**		3	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	0	0	45
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Describe cyber laws, IT Act, data protection and various cybercrimes.	U	1,6,8
2	Analyze and apply security measures during online transactions and financial activities.	An	1,2,10
3	Illustrate basic cryptographic techniques and importance of cyber forensic.	U	2,3,6,10

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S),Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	0	0	0	0	3	0	2	0	0
CO 2	2	3	0	0	0	0	0	0	0	2
CO 3	0	3	2	0	0	2	0	0	0	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level)

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I - Cyber Laws, IT Act and Cyber Crimes (20 Hours)**			
	1.1	Introduction to Cyber laws: Definition and Scope, Key legal concepts in cyberspace.	2	1
	1.2	IT Act: Overview of the IT Act 2000, Offenses and penalties under the IT Act, Amendments and evolving landscape.	4	1
	1.3	Data Protection and Privacy Laws : Principles of Data Protection, Privacy laws and regulations.	3	1
	1.4	Cyber Crimes: Types of Cybercrimes, Hacking and unauthorized access, Identity theft and cyber fraud.	4	1
	1.5	Cyber Bullying and Harassment: Definition and Forms of Cyber Bullying, Legal Perspective on Cyberbullying.	4	1
	1.6	Harassment Laws and social media, Reporting and preventing cyberbullying.	3	1
2	**Module II - Online Security (11 Hours)**			
	2.1	Introduction to Internet Security: Overview of Internet Security, Importance of Online Safety.	2	2
	2.2	Passwords and Authentication: Importance of Strong Password, Multi Factor Authentication (MFA).	2	2

	2.3	Secure Browsing Practices: Recognizing and Avoiding phishing Attacks, Identifying Secure Websites (HTTPS).	3	2
	2.4	Social Media Security: Privacy settings on Social media platforms, Secure sharing information.	2	2
	2.5	Online Transaction and Financial Security: Secure online shopping, Banking and Financial Security, Payment Card safety.	2	2
3	**Module III - Introduction to Cryptography and Cyber Forensics (14 Hours)**			
	3.1	Security Concepts: Introduction, The need for security, Principles of security, Types of Security attacks	3	3
	3.2	Cryptography Concepts and Techniques: Introduction, plain text and cipher text, substitution techniques, transposition techniques,	4	3
	3.3	Encryption and decryption, symmetric and asymmetric key Cryptography	3	3
	3.4	Introduction to Cyber forensics - Definition and importance of cyber forensics, Types of cybercrime -hacking, phishing, identity theft, etc., The role of forensics in investigating cybercrime. Introduction to Ethical Hacking.	4	3
4	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	**MODE OF ASSESSMENT** **A. Continuous Comprehensive Assessment (CCA)** **CCA for Theory(25 Marks)** 1. Written Test 2. Assignments 3. Seminars **B. End Semester Evaluation (ESE) – Non-MCQ**

	ESE for Theory : Written Test (50 Marks, 1.5 hrs) Part A : Very Short Answer Questions(10 out of 12 Questions) - (10*2=20 Marks) Part B : Short Answer Question (4 out of 6 Questions) – (4*5=20 Marks) Part C : Essay Questions(1 out of 3 Questions) – (1*10=10 Marks)

REFERENCES:

1. Vakul Sharma, “Information Technology Law and Practice”, 3rd ed. 2011, Universal Law Pub., New Delhi.
2. Adv. Prashant Mali, “Cyber Law & Cyber Crimes”, Snow White Publications Pvt. Ltd, 2nd ed. 2015.
3. Michael Cross , “Social Media Security: Leveraging Social Networking While Mitigating Risk”, Elsevier, 2014.

SUGGESTED READINGS:

1. William Stallings & Lawrie Brown “ Computer Security – Principles and Practice” 3rd ed., Pearson Pub., 2017.
2. William Stallings, Cryptography and Network Security Principles and Practice, 4/e,Pearson Ed.
3. Cyber Forensics - Concepts and Approaches, Ravi Kumar & B Jain,2006, icfai university press

SEMESTER VI

Semester: 6

Course Code	Title of the Course	Type of the Course	Credit	Hours/ Week	Hour Distribution /week		
					L	P	O
MG6SDCSDS300	Mobile Application Development	SDC	4	5	3	2	0
MG6SDESDS300	Java Programming	SDE(Elective Papers - Select one)	4	5	3	2	0
MG6SDESDS301	Data Analytics Using Python						
MG6SECSDS300	Big Data Analytics	SEC	3	3	3	0	0
MG6MPCSDS300	Introduction to Machine Learning	MPC	4	4	4	0	0
MG6VACSDS300	Methodologies for Computational Research	VAC	3	3	3	0	0
MG6PRJSDS300	Project	PRJ	4	8	0	8	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

(logo)	# Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**Mobile Applications Development**
Type of Course	**SDC**
Course Code	**MG6SDCSDS300**
Course Level	**300**
Course Summary	This course introduces students to mobile app development on the Android platform (Android Studio) with a focus on integrating SQLite for local data storage. Learners will gain hands-on experience in designing user interfaces and implementing operations to manage persistent data within Android applications. By the end of the course, students will be able to develop functional, database-driven mobile apps using SQLite.

Semester	6	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if Any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the evolution, versions, and basic architecture of Android OS. Identify the development environment components such as JDK, SDK, and AVD. Apply and differentiate various layout types in Android development.	U	1,2,10

2	Develop intuitive and interactive UI using Android Views and WidgetsCO2. Customize UI elements based on user interaction. Implement dynamic and complex user interfaces with Android controls.	S	1,2,3,5,9,10
3	Explain the working and lifecycle of Android activities and components. Implement multimedia services and system resources in Android. Demonstrate inter-component communication using intents and broadcasts.	An	1,2,3,5,8,9,10
4	Integrate SQLite database in Android apps for persistent storage. Create, read, and manipulate data using cursors and transactions. Build apps with messaging and connectivity features. Interpret and use JSON and XML for data interchange in Android. Integrate Google Play and Location services into applications. Handle real-time data and display it with mapping APIs.	C	1,2,3,5,6,8,9,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**2**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 2	**2**	**3**	**2**	**0**	**1**	**0**	**0**	**0**	**1**	**3**
CO 3	**3**	**3**	**2**	**0**	**1**	**0**	**0**	**1**	**1**	**3**
CO 4	**3**	**3**	**2**	**0**	**1**	**1**	**0**	**1**	**2**	**3**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
	Module I - Android Application Development Fundamentals (14 Hours)			
1	**1.1**	Introduction to Android, Android Versions, Android Activity, Android Features and Architecture,	4	1
	1.2	Java JDK, Android SDK, Android Development Tools, Android Virtual Devices, Emulators,	5	1

	1.3	Dalvik Virtual Machine, Layouts – Linear, Absolute, Frame, Relative and Table.	5	1
2	**Module II - Android UI Components and Design (14 Hours)**			
	2.1	Android User Interface- Fundamental UI design , User interface with View- Text View,	5	2
	2.2	Buttons, Image Button, Edit Text, Check Box, Toggle Button, Radio Button and Radio Group, Progress Bar, Autocomplete	5	2
	2.3	Text View, Spinner, List View, Grid View, Image View, Scroll View, Custom Toast Alert and Time and Date Picker..	4	2
3	**Module III - Android Services, Multimedia, and Data Management (17 Hours)**			
	3.1	Activity - Introduction, Intent, Intent_filter, Activity Life Cycle, Broadcast Life Cycle, Services,	3	3
	3.2	multimedia-Android System Architecture, Play Audio and Video, Text to Speech.	4	3
	3.3	SQLite Database in Android- Introduction to SQLite Database, Creation and Connection of the Database, Extracting values from Cursors, Transactions, Telephoning and Messaging-SMS Telephony, Sending SMS, Receiving SMS, Wi-Fi Activity.	5	3
	3.4	Introduction to JSON and XML, Use of JSON, Syntax and Rule of JSON, JSON Name, JSON Values, JSON Objects, JSON Arrays, Parsing JSON and XML. Google Play services, Location services, Maps.	5	3
	Module - IV *(Practical Session)*			

4	**4.1**	1. SQLite Database: Introduction, Connection, Transactions, Cursors 2. Telephony & Messaging: SMS Send/Receive, Telephony Manager 3. Connectivity: Wi-Fi Activity 4. JSON & XML: Syntax, Rules, Parsing, Objects, Arrays 5. Google Play Services: Location Services, Maps API	10	4
	4.2	Android Lab Questions Based on SQLite Database 1. Create a simple Android app that demonstrates creating a SQLite database and a single table. 2. Insert user details into a SQLite database using Edit Text fields. 3. Display all saved records from SQLite in a List View. 4. Update,delete,Search,an existing record based on user input. 5. Validate input fields before inserting into the database. 6. Create a login form that checks credentials from the database. 7. Display data from SQLite using Recycler View. 8. Student Record App with Add, Edit, View, Delete features.	20	4
5	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
	MODE OF ASSESSMENT **Mode of Assessment**

Assessment Types	**A. Continuous Comprehensive Assessment (CCA)** **CCA for Practical (30 Marks)** **1.** Lab Assessment 2. Observation of Practical Skills 3. Viva
	B. End Semester Evaluation (ESE) **ESE for Practical : Practical Examination(70 Marks,3 hrs)** Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES

1. Kevin Grant and Chris Haseman, Beginning Android Programming – Develop and Design, Pearson
2. Meier, R. (2012). Professional Android 4 application development (3rd ed.). Wrox.
3. Haseman, C., & Murphy, K. (2011). Android programming (1st ed.). McGraw-Hill Education.
4. Darcey, L., & Conder, S. (2012). Android wireless application development (3rd ed.). Addison-Wesley.
5.

SUGGESTED READINGS

1. Prasanna Kumar Dixit - ANDROID, Vikas Publishing House.
2. AnubhavPradhan, Anil Deshpande, Composing Mobile Apps using Android, Wiley India Pvt.Ltd,2014

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**				
Programme	**B.Voc.(Honours) Software Development and System Administration**				
Course Name	**Java Programming**				
Type of Course	**SDE**				
Course Code	**MG6SDESDS300**				
Course Level	**300**				
Course Summary	This course provides a foundational understanding of object-oriented programming using Java, covering core concepts such as syntax, control structures, classes and objects, inheritance, and exception handling. Emphasizing hands-on practice, it equips students to build modular, reusable, and maintainable Java applications, laying the groundwork for advanced programming and real-world software development.				
Semester	6	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Apply OOP concepts and Java fundamentals to develop robust programs.	A	1,2,10
2	Analyze class structure, inheritance, method implementation, and array handling in Java.	An	1,2

3	Demonstrate Java packages, exception handling, multithreading, Swing components, event handling and database connectivity.	A	2,3,10
4	Demonstrate proficiency in Java programming through practical implementation and problem- solving.	A	2,4,10

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	3	0	0	0	0	0	0	0	2
CO 2	2	3	0	0	0	0	0	0	0	0
CO 3	0	3	2	0	0	0	0	0	0	3
CO 4	0	3	0	2	0	0	0	0	0	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CONo.
Upon completion of this course students will able to understand:				
1	**Module I - Java Fundamentals and Object-Oriented Concepts (10 Hours)**			
	1.1	Concepts of Object Oriented Programming, Benefits of OOP,	3	1
	1.2	Features of Java, Java Environment, Java tokens. Constants, variables, data types, operators.	3	1
	1.3	Control statements-branching, looping and jump statements, labelled loops.	4	1
2	**Module II – Java Classes, Inheritance, and Arrays (17 hours)**			

	2.1	Defining a class, fields declaration, method declaration , creating object, accessing class members	3	2
	2.2	Method overloading, constructors, constructor overloading,	3	2
	2.3	Command line arguments, super keyword, static members,	3	2
	2.4	Inheritance, overriding methods, dynamic method despatch, final(variables, methods and classes), abstract methods and classes, interfaces, visibility control.	4	2
	2.5	Arrays-one dimensional arrays, declaration, creation, initialization of arrays, two dimensional arrays. String class.	4	2
3	**Module III - Java Packages, Multithreading, and GUI Programming (18 Hours)**			
	3.1	Packages:- Java API packages overview(lang, util, io, swing, applet), user defined packages-creating packages, using packages.	4	3
	3.2	Exception handling techniques, Multithreading- creation of multithreaded program-Thread class – Runnable interface-thread life cycle.	3	3
	3.3	Swing components-ImageIcon, JLabel, JTextField, JTextArea, JButton, JCheckBox, JRadioButton, JList, JComboBox, JTable, JTabbedPane, JScrollPane,	4	3
	3.4	Event handling –Delegation Event Model-event classes-sources of events-evenlisteners.	3	3
	3.5	**Database Connectivity :** JDBC architecture-JDBC connection, JDBC statement object, JDBC drivers.	4	3

4	**4.1**	**(practical session)** 1. Implement basic OOP concepts through hands-on exercises. 2. Develop Java applications demonstrating inheritance and polymorphism 3. Utilize arrays and strings in practical coding tasks. 4. Create and use packages 5. Implement exception handling techniques 6. Build multithreaded Java programs to handle concurrent tasks efficiently. 7. Design and develop graphical user interfaces using Swing components. 8. Implement event handling programs	30	4
5		**(Teacher Specific Content)**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Use of ICT tools in conjunction with traditional classroom teaching methods ● Interactive sessions ● Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment**
	A. Continuous Comprehensive Assessment (CCA) **CCA for Practical (30 Marks)** 1. Lab Assessment 2. Observation of Practical Skills 3. Viva
	B. End Semester Evaluation (ESE)

	ESE for Practical : Practical Examination(70 Marks,3 hrs) Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record:10 Marks

REFERENCES

1. Balagurusamy (2014). Programming with Java (3rd Edition). McGraw Hill Education.
2. Patrick Naughton (2002). Java 2 The Complete Reference (7th Edition). Osborne/McGraw-Hill

SUGGESTED READINGS

1. Horstmann, C. S., & Cornell, G. (2008). *Core Java, Volume 1: Fundamentals* (8th ed.). Prentice Hall.
2. Somasundaram, K. (Year). *Programming in Java 2* (1st ed.). Jaico Publishing House

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/Software Development and System Administration**					
Programme	**B.Voc.(Honours) Software Development and System Administration**					
Course Name	**Data Analytics Using Python**					
Type of Course	**SDE**					
Course Code	**MG6SDESDS301**					
Course Level	**300**					
Course Summary	This course is designed to teach students how to analyze different types of data using Python. Students will learn how to prepare data for analysis, perform simple statistical analysis, create meaningful data visualizations and predict future trends from data.					
Semester	6	**Credits**			4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	**Others**	
		3	1	0	0	75
Pre-requisites, if any	**NIL**					

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Describe data, its type and the data analysis process.	U	1,2,10
2	Illustrate the numerical computation of data using NumPy library.	U	1,2
3	Analyse data manipulation and visualization using Pandas and Matplotlib library respectively.	An	1,2
4	Implement various Python libraries to perform data analysis tasks.	A	2,4,8,10

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	0	0	0	0	0	0	0	2
CO 2	3	3	0	0	0	0	0	0	0	0
CO 3	3	3	0	0	0	0	0	0	0	0
CO 4	0	3	0	2	0	0	0	3	0	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
	Module I - Introduction to Data analysis (10 Hours)			
1	1.1	Understanding structured and unstructured data	2	1
	1.2	Data Analysis process -Defining objectives, Data collection, Data cleaning, Data analysis, Data interpretation and visualization, Types of data Analysis	8	1
	Module II - Numerical Computation with NumPy Library (15 Hours)			
2	2.1	Ndarray, Creating Ndarrays, Data types for Ndarrays, Arithmetic with NumPy Arrays	5	2
	2.2	Basic Indexing and Slicing, Boolean Indexing and Fancy indexing.	5	2
	2.3	Universal functions, Mathematical and statistical functions, Sorting.	5	2
	Module III - Data Manipulation and Data visualization (20 Hours)			

3	**3.1**	**Data Manipulation with Pandas Library**: Introduction to Pandas Object, Pandas data structures- Series, DataFrame, Index object.	6	3
	3.2	Functionalities- Reindexing, Indexing, Selection, Filtering, Sorting, Ranking, Summarizing and Computing Descriptive Statistics, Handling missing data, Hierarchical indexing.	6	3
	3.2	**Introduction to Data visualization**: Matplotlib Library, pyplot, Data visualization using matplotlib - bar plot, line plot, histogram, pie chart, box plots, density plots and scatter plot.	8	3
4	**(Practical Session)**			
	4.1	Implementation of ND array Basic Operations - Indexing, Slicing and Iterating, Conditions and Boolean Arrays, Shape Manipulation, Array Manipulation	10	4
	4.2	Implementation of Data Frames, Reading from csv files, Python programs to use data cleaning, loc() function, iloc() function, Descriptive Statistics – count(), sum(), mean(), median(), mode(), std(), min(), max() and cumsum(). Reading from csv files, Data cleaning, Inserting columns into Data Frames, Deleting columns from Data Frame, Concatenating Data Frame, Writing back to csv files.	10	4
	4.3	Data Visualization using bar plot, line plot, histogram, pie chart, box plots, density plots, and scatter plot.	10	4
5	**Teacher Specific Content**			

<table>
<tr><td>Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)
Use of ICT tools in conjunction with traditional classroom teaching methods
Interactive sessions
Class discussions</td></tr>
<tr><td rowspan="3">Assessment Types</td><td>MODE OF ASSESSMENT

Mode of Assessment</td></tr>
<tr><td>A. Continuous Comprehensive Assessment (CCA)

CCA for Practical (30 Marks)
Lab Assessment
Observation of Practical Skills
Viva</td></tr>
<tr><td>B. End Semester Evaluation (ESE)

ESE for Practical : Practical Examination(70 Marks,3 hrs)
Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks)
Part B : Practical Questions(2 Questions) – (2*15=30 Marks)
Viva : 10 Marks
Record : 10 Marks</td></tr>
</table>

REFERENCES

1. Wes Mckinney, “Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter” 3rEdition, O'Reilly,2022.Free online access: https://wesmckinney.com/book/
2. VanderPlas, J. (2016). Python data science handbook: Essential tools for working with data. O’Reilly Media.

SUGGESTED READINGS

1. Fabio Nelli, "Python Data Analytics Data Analysis and Science Using Pandas, Matplotlib, and the Python Programming Language", Edition 1, 2015, Apress.
2. William McKinney, "Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Ipython", Edition 2, 2017, Shroff/O'Reilly.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**Big Data Analytics**
Type of Course	**SEC**
Course Code	**MG6SECSDS300**
Course Level	**300**
Course Summary	This course introduces Big Data and Data Analytics techniques, covering the architecture, methods, and modern tools like Hadoop, Spark, and NoSQL. It covers data analytics lifecycle, machine learning concepts, and applications in finance, healthcare, retail, and IoT. Students will learn to apply big data tools for real-world problem-solving and intelligent business decisions.

Semester	6	**Credits**		3	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	0	0	45
Pre-requisites, if Any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the fundamentals, characteristics, and architecture of big data systems.	U	1,2,3,6,9,10
2	Apply analytical techniques for processing and interpreting large-scale data.	A	1,2,3,6,8,9,10
3	Analyze real-time data using big data tools and platforms. Evaluate and develop big data solutions for business intelligence and decision-making.	An	1,2,3,5,6,8,9,10

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**2**	**2**	**2**	**0**	**0**	**1**	**0**	**0**	**1**	**3**
CO 2	**3**	**3**	**2**	**0**	**0**	**1**	**0**	**1**	**1**	**3**
CO 3	**3**	**3**	**3**	**0**	**1**	**1**	**0**	**1**	**2**	**3**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module 1: Introduction to Big Data(15 Hours)**			
	1.1	Evolution of Big Data	2	1
	1.2	Characteristics of Big Data: Volume, Variety, Velocity, Veracity, and Value (5Vs)	3	1
	1.3	Traditional vs Big Data System	2	1
	1.4	Big Data Architecture and Ecosystem	2	1
	1.5	Storage Models: HDFS, NoSQL (MongoDB, Cassandra)	2	1
	1.6	Introduction to Distributed Computing (Hadoop & MapReduce)	2	1
	1.7	Cloud and Big Data (AWS, Azure, GCP overview)	2	1
2	**Module 2: Data Analytics Techniques (15 Hours)**			
	2.1	Data Preprocessing and Cleaning	3	2
	2.2	Types of Analytics: Descriptive, Diagnostic, Predictive, Prescriptive	3	2
	2.3	Statistical Analysis and Data Exploration	3	2

	2.4	Data Visualization Tools: Tableau, Power BI, Matplotlib	3	2
	2.5	Introduction to Machine Learning in Analytics-Regression, Classification, Clustering	3	2
3	**Module 3: Tools, Platforms & Applications of Big Data Analytics (15 Hours)**			
	3.1	Big Data Platforms: Hadoop, Spark, Hive, Pig, Flink	3	3
	3.2	Data Warehousing and ETL concepts	2	3
	3.3	Data Governance, Privacy, and Security in Big Data	3	3
	3.4	Business Intelligence and Decision Support Systems	2	3
	3.5	Applications in Finance, Healthcare, Retail, and IoT	3	3
	3.6	Emerging Trends: Edge Analytics, AI integration, DataOps	2	3
4	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment**
	A. Continuous Comprehensive Assessment (CCA) **CCA for Theory(25 Marks)** 1. Written Test 2. Assignment 3. Seminar

	B. End Semester Evaluation (ESE) – Non-MCQ **ESE for Theory : Written Test(50 Marks,1.5 hrs)** Part A : Very Short Answer Questions(10 out of 12 Questions) – (10*2=20 Marks) Part B : Short Answer Questions(4 out of 6 Questions) - (4*5=20 Marks) Part C : Essay Questions(1 out of 3 Questions) - (1*10=10 Marks)

REFERENCES

1. Mayer-Schönberger, V., & Cukier, K. (2013). Big data: A revolution that will transform how we live, work, and think. Houghton Mifflin Harcourt.
2. Minelli, M., Chambers, M., & Dhiraj, A. (2013). Big data, big analytics: Emerging business intelligence and analytic trends for today's businesses. Wiley.

SUGGESTED READINGS

1. White, T. (2015). Hadoop: The definitive guide (4th ed.). O'Reilly Media. 2)George, L. (2011). HBase: The definitive guide. O'Reilly Media.
2. Russell, M. (2011). Mining the social web: Analyzing data from Facebook, Twitter, LinkedIn, and other social media sites. O'Reilly Media.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**				
Programme	**B.Voc.(Honours) Software Development and System Administration**				
Course Name	**Introduction to Machine Learning**				
Type of Course	**MPC**				
Course Code	**MG6MPCSDS300**				
Course Level	**300**				
Course Summary	This course provides students with a comprehensive understanding of Machine Learning (ML) concepts, methods, and tools, covering supervised and unsupervised learning techniques, real-world applications, and Python libraries.				
Semester	6	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		4	0	0	60
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Define key concepts, tools, and workflows in machine learning	U	1,2
2	Analyze supervised learning techniques to solve	An	1,2,3

	classification/regression tasks		
3	Implement unsupervised learning methods like clustering and PCA	An	1,2,3,4,10
4	Evaluate and improve ML models using various techniques	An	1,2,3,4,5,9,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/P O	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	3	0	0	0	0	0	0	0	0
CO 2	2	3	2	0	0	0	0	0	0	0
CO 3	2	3	2	2	0	0	0	0	0	3
CO 4	3	3	2	2	2	0	0	0	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level)

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I: Foundations of Machine Learning (15 Hours)**			
	1.1	Introduction to AI and ML: definitions, types of ML	3	1
	1.2	Key ML concepts: features, labels, overfitting, underfitting	3	1
	1.3	Applications of ML across industries	3	1
	1.4	The ML pipeline: data collection, preprocessing, training, evaluation	3	1

	1.5	Python libraries for ML: NumPy, pandas, matplotlib, scikit-learn	3	1
2	**Module II: Supervised Learning (15 Hours)**			
	2.1	Regression vs. classification	2	2
	2.2	Linear Regression: hypothesis, cost function, gradient descent	3	2
	2.3	Logistic Regression	2	2
	2.4	k-Nearest Neighbors (k-NN)	3	2
	2.5	Decision Trees and Random Forest	2	2
	2.6	Evaluation Metrics: Accuracy, Precision, Recall, F1-Score	3	2
3	**Module III: Unsupervised Learning (15 Hours)**			
	3.1	Clustering: k-Means Clustering, Hierarchical Clustering	4	3
	3.2	Dimensionality Reduction: PCA	3	3
	3.3	Anomaly Detection	4	3
	3.4	Real-life examples of unsupervised learning	4	3
4	**Module IV: Model Evaluation and Real-World Implementation (15 Hours)**			
	4.1	Data Preprocessing: normalization, encoding, missing values	3	4

	4.2	Train/test split, cross-validation	3	4
	4.3	Bias-variance tradeoff	3	4
	4.4	Hyperparameter tuning	3	4
	4.5	Introduction to model deployment concepts	3	4
5	**Teacher Specific Content**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment**
	A. Continuous Comprehensive Assessment (CCA) **CCA for Theory(30 Marks)** 1. Written Test 2. Problem Based Assignment 3. Seminars **B. End Semester Evaluation (ESE) – Non-MCQ** **ESE for Theory : Written Test(70 Marks , 2 hrs)** Part A: Very Short Answer Questions (10 out of 12 Questions) - (10*2=20 Marks) Part B: Short Answer Questions (4 out of 6 Questions) - (4*5=20 Marks) Part C: Essay Questions (2 out of 4 Questions) - (2*15=30 Marks)

REFERENCES

1. Géron, A. (2019). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems (2nd ed.). O'Reilly Media.
2. Alpaydin, E. (2020). Introduction to Machine Learning (4th ed.). MIT Press.

SUGGESTED READINGS

1. Aggarwal, C. C. (2018). Neural Networks and Deep Learning: A Textbook. Springer.
2. Shalev-Shwartz, S., & Ben-David, S. (2014). Understanding Machine Learning: From Theory to Algorithms. Cambridge University Press.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**Methodologies for Computational Research**
Type of Course	**VAC**
Course Code	**MG6VACSDS300**
Course Level	**300**
Course Summary	This course offers a foundation in research principles, data analysis, and computing research development. It builds analytical skills through practical use of parametric tests and explores ethical issues like plagiarism and data misuse. Students gain the tools for responsible and effective research.

Semester	6	**Credits**		3	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	0	0	45
Pre-requisites, if any	**NIL**				

COURSE OUTCOMES (CO)

CONo	Expected Course Outcome	Learning Domains	PO No
1	Describe research methodology, including objectives, types, approaches, and significance.	U,	1,2,10
2	Explain the comprehensive framework of research methodology and scientific method.	U	1,2,10

3	Describe the historical evolution of computing ideas, explore research methods, and analyze the application of computers in research.	,An	1,2,3,6,10

Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	0	0	0	0	0	0	0	2
CO 2	2	3	0	0	0	0	0	0	0	2
CO 3	2	2	3	0	0	3	0	0	0	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
Upon completion of this course students will able to understand:				
1	**Module I : Fundamentals of Research and Data Analysis (10 Hours)**			
	1.1	Meaning, Objectives, Motivation, Types, Approaches and Significance of Research	2	1
	1.2	Reading and Reviewing-Research literature, Finding Research Papers, Criteria of Good Research.	2	1
	1.3	Data analysis in Research: Introduction, Need for Data Collection, Methods of Data Collection	3	2
	1.4	Principles for Accessing Research Data, Data Processing, Data Analysis, Presentation of Data,	3	2
2	**Module II : Research Methods, Statistical Analysis, and Reporting (15 hours)**			
	2.1	Error Analysis, Scientific Models. Scientific Methodology - Introduction Rules and Principles of Scientific Method.	3	2
	2.2	Hypothesis Testing of Hypothesis, Basic concepts, Procedure, Important parametric tests: z-test ,t-test, $\chi 2$ - square test, F test	4	2

	2.3	Ethics in Research, Technical Reports- Bibliography referencing and footnotes.	4	2
	2.4	Research in Practice- Literature Review, Journals, Conference Proceedings, journal Impact Factor, citation Index, h Index.	4	2
	Module III : Computing Research Methods and Applications (15 hours)			
3	**3.1**	History of ideas in computing, Evolution of Computing Research	3	3
	3.2	Overview of Research Methods: Measurements based research methods - Deductive Methods - Inductive Methods.	4	3
	3.3	The significance of Interdisciplinary research for Computer Science.	4	3
	3.4	Application of Computer in Research --MS office and its application in Research, Use of Internet in Research – Websites, search Engines, E-journal and E-Library.	4	3
4	**(Teacher specific content)**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	**MODE OF ASSESSMENT** **Mode of Assessment**
	A. Continuous Comprehensive Assessment (CCA) **CCA for Theory(25 Marks)** 1. Written Test 2. Assignment 3. Paper Presentation

	B. End Semester Evaluation (ESE) – Non-MCQ **ESE for Theory : Written Test(50 Marks,1.5 hrs)** Part A : Very Short Answer Questions(10 out of 12 Questions) – (10*2=20 Marks) Part B : Short Answer Questions(4 out of 6 Questions) - (4*5=20 Marks) Part C : Essay Questions (1 out of 3 Questions) - (1*10=10 Marks)

REFERENCES

1. Kothari, C.R.(1990). Research Methodology: Methods and Techniques. New Age International. Publishers(Second revised edition).
2. Saunders, M., Lewis, P., & Thornhill, A. (2019). Research methods for business students (8th ed.). Pearson Education.
3. Kumar, R. (2019). Research methodology: A step-by-step guide for beginners (5th ed.). SAGE Publications.

SUGGESTED READINGS

1. Krishnan Nallaperumal, "Engineering Research Methodology : A Computer Science and Engineering and Information and Communication Technologies Perspective. " (First Edition)
2. Justin Zobel,Writing For Computer Science, Springer (Third Edition)
3. K Prathapan,Research Methodology for Scientific Writing ,I.K International Publishing House Pvt.Ltd
4. S.P Satarkar, S.V., 2000. Intellectual Property Rights and Copy right. Ess Publication

	# Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**				
Programme	**B.Voc.(Honours) Software Development and System Administration**				
Course Name	**Project**				
Type of Course	**PRJ**				
Course Code	**MG6PRJSDS300**				
Course Summary	The project work provides students with an opportunity to identify, analyze, and solve real-world problems relevant to their field of study by integrating and applying the theoretical knowledge and skills acquired throughout their academic program. It fosters independent research, critical thinking, innovation, and the practical use of methodologies, tools, and techniques to design effective solutions. Students are encouraged to work individually or in teams, enhancing their collaboration, time management, ethical responsibility, and self-directed learning. The project also develops competencies in academic writing, documentation, and technical communication. Each project is expected to culminate in a comprehensive report, a working model or prototype (where applicable), and a formal presentation followed by a viva voce examination, demonstrating the student's ability to apply knowledge creatively and professionally in a real-world context.				
Semester	**6**	**Duration**	**8 hours/week**	**Credits**	**4**

COURSE OUTCOMES (CO)

CO No:	Expected Course Outcome	Learning Domains	PO No:
	Upon the successful completion of the course, the student will be able to		

1	Identify, analyze, and define problems relevant to the field of study.	An	1,2,3
2	Apply appropriate methodologies, tools, and techniques to design and implement effective solutions.	C	2,3,10
3	Demonstrate skills in research, critical thinking, project planning, and systematic execution.	S	1,2,5,10
4	Produce well-structured academic reports and communicate project outcomes effectively.	S	4,8,10
5	Exhibit teamwork, time management, ethical responsibility, and initiative in a self-directed project environment.	S	5,8,9,10
6	Address real-world challenges with innovative and context-aware solutions.	Ap	1,2,6,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	3	1	0	0	0	0	0	0	0
CO 2	0	3	2	0	0	0	0	0	0	2
CO 3	3	3	0	0	2	0	0	0	0	3
CO 4	0	0	0	2	0	0	0	3	0	3
CO 5	0	0	0	0	3	0	0	2	1	1
CO 6	3	2	0	0	0	1	0	0	0	1

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

Assessment Types	**MODE OF ASSESSMENT**		
	A	**Continuous Comprehensive Assessment (CCA)**	
		Components	**Marks**
		Commitment and Involvement	5
		Periodic progress review	10
		Quality of work/Implementation effort	10
		Report	5
		Total	**30**
	B	**End Semester Evaluation (ESE)**	

		Components	Marks
		Problem Identification and Objectives	10
		Methodology / Design / Technical Content	15
		Implementation / Analysis / Results	15
		Final Report	10
		Presentation	10
		Viva Voce	10
		Total	**70**

SEMESTER: VII & VIII

Semesters: 7 & 8

B.Voc. Honours							
Course Code	**Title of the Course**	**Type of the course**	**Credit**	**Number of Days**	**Credit Distribution**		
					L	**P**	**O**
MG7APPSDS400	Apprenticeship	APP	28	280 days	0	28	0
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			

B.Voc. Honours with Research							
Course Code	**Title of the course**	**Type of the course**	**Credit**	**Number of Days**	**Credit Distribution**		
					L	**P**	**O**
MG7RINSDS400	Research Internship	RIN	20	200 days	0	20	0
NA	Online	SDC	4	NA			
NA	Online	SDC	4	NA			
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			

L — Lecture, P — Practical/Practicum , O — On the Job Training

	Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**Apprenticeship**
Type of Course	**APP**
Course Code	**MG7APPSDS400**
Summary	As an integral component of the B.Voc. Honours degree programme, students are required to complete a structured apprenticeship or work-integrated learning programme in collaboration with relevant industries, organizations, or institutions. This component, spanning a duration of 280 days, carries 28 academic credits and is compulsory in the student's designated skill domain. It is designed to enhance industry preparedness by reinforcing academic knowledge through sustained, domain-relevant practical experience. The apprenticeship offers students the opportunity to engage directly with real-world professional environments, enabling them to apply domain-specific competencies, gain exposure to industry-standard tools and practices, and participate meaningfully in ongoing operations and projects. This extended, immersive experience serves to bridge the gap between theoretical learning and professional expectations, thereby fostering critical skills for career development and employability. To ensure the effectiveness, academic relevance, and accountability of the apprenticeship: • Each student will be assigned an academic mentor from the parent institution and an industry supervisor from the host organization. • Students are required to maintain a weekly activity logbook, which must be regularly reviewed and signed by the industry supervisor. • Monthly progress reports will be submitted to and reviewed by the academic mentor in consultation with the industry supervisor.

	• Mid-term and final evaluations will be conducted based on a combination of employer feedback, student outputs/deliverables, and academic performance metrics. • The institution will conduct site visits, virtual check-ins, or regular follow-upsto ensure student engagement, address issues promptly, and uphold the quality of the apprenticeship experience. This structured apprenticeship is a critical step in preparing students for the dynamic demands of the professional world, ensuring that their academic journey culminates ina well-rounded and industry-aligned skill set.				
Semester	**7&8**	**Duration**	**280 days**	**Credits**	**28**

COURSE OUTCOMES (CO)

CO No:	Expected Course Outcome	Learning Domains	PO No:
Upon the successful completion of the course, the student will be able to:			
1	Gain hands-on professional experience by engaging in long-term, domain-specific apprenticeship in real-world industry environments.	S	1,3,6,10
2	Apply domain-specific theoretical knowledge to solve real-time problems, enhancing technical and problem-solving competencies.	A	1,2,3,10
3	Demonstrate professional competencies such as workplace etiquette, communication skills, and teamwork in a collaborative work culture.	S	4,5,8,9
4	Build a professional portfolio by achieving practical outcomes and establishing credible industry references and credentials.	C	5,9,10
5	Cultivate reflective thinking, adaptability, and a lifelong learning mindset through structured and mentored work experience.	Ap	1,6,8,10
6	Transition smoothly from academic study to professional practice by developing job-specific skills and industry-aligned competencies.	S	2,3,5,10
***Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)**			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**3**	**0**	**3**	**0**	**0**	**2**	**0**	**0**	**0**	**2**

CO 2	**3**	**2**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 3	**0**	**0**	**0**	**3**	**2**	**0**	**0**	**1**	**2**	**0**
CO 4	**0**	**0**	**0**	**0**	**3**	**0**	**0**	**0**	**2**	**3**
CO 5	**3**	**0**	**0**	**0**	**0**	**2**	**0**	**2**	**0**	**1**
CO 6	**0**	**3**	**3**	**0**	**2**	**0**	**0**	**0**	**0**	**1**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

Assessment Types	**MODE OF ASSESSMENT**		
	A	**Continuous Comprehensive Assessment(CCA)**	
		Components	**Marks**
		Commitment, Punctuality & Professional Conduct	20
		Monthly Progress Reviews & Logbook Maintenance	25
		Skill Development & Application	25
		Interim Report	20
		Total	**90**
	B	**End Semester Evaluation(ESE)**	
		Components	**Marks**
		Feedback & Evaluation Report from Host Organization	50
		Skill Demonstration / Summary of Work Exposure	40
		Final Report / Learning Portfolio	40
		Domain Knowledge and Experience Communication (Presentation)	40
		Viva Voce	40
		Total	**210**

Note:

This assessment framework is intended as a guiding structure for evaluating apprenticeship performance. However, in order to remain responsive to the evolving needs of industry and society, the evaluation criteria may be revised from time to time. Such changes aim to enhance the relevance, effectiveness, and fairness of the assessment process.

	# Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**				
Programme	**B.Voc.(Honours) Software Development and System Administration**				
Course Name	**Research Internship**				
Type of Course	**RIN**				
Course Code	**MG7RINSDS400**				
Summary	As an integral requirement of the B.Voc. Honours with Research degree programme, the Research Internship is designed to provide students with hands-on exposure to real- world research practices in their designated skill domain. This component carries 20 academic credits and extends over a duration of 200 days. The internship must be undertaken in collaboration with a research organization, industry, or university department, under the mentorship of a qualified research guide. The primary aim of this internship is to engage students in industry-linked research projects that allow them to apply theoretical knowledge to practical, domain-specific problems. Students are expected to work on meaningful research inquiries, contribute to data collection and analysis, develop critical thinking and problem-solving skills, and enhance their communication and documentation abilities. In addition to the research internship, students must earn 8 credits through Skill Development Courses (SDCs), specifically chosen for their research orientation, thereby reinforcing their academic and practical foundation. This component not only contributes significantly to the academic rigor of the Honours with Research degree but also ensures a seamless transition from classroom learning to workplace research, preparing students for advanced studies or professional roles in their respective domains.				
Semester	**7&8**	**Duration**	**200 days**	**Credits**	**20**

COURSE OUTCOMES (CO)

CO No:	Expected Course Outcome	Learning Domains	PO No:
Upon the successful completion of the course, the student will be able to:			
1	Demonstrate research aptitude and inquiry-based learning by actively engaging in real-time research projects.	S	1,2,10
2	Apply academic knowledge in a professional research environment to bridge the gap between theory and real-world research practices.	A	2,3,6,10
3	Strengthen domain-specific knowledge and technical competencies through systematic investigation and practical application.	S	1,2,3
4	Address real-world research problems using problem-solving, analytical, and critical thinking skills.	S	1,2,6
5	Communicate scientific ideas and findings effectively through research reports, documentation, and presentations.	S	4,8,10
6	Collaborate with researchers and peer groups to gain exposure to interdisciplinary perspectives and collaborative learning practices.	S	2,3,5,10
7	Demonstrate professional growth and readiness for higher education, entrepreneurship, or research-oriented careers.	I	5,9,10
***Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)**			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**3**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 2	**0**	**3**	**2**	**0**	**0**	**2**	**0**	**0**	**0**	**1**
CO 3	**3**	**2**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
CO 4	**3**	**2**	**0**	**0**	**0**	**2**	**0**	**0**	**0**	**0**
CO 5	**0**	**0**	**0**	**3**	**0**	**0**	**0**	**2**	**0**	**1**
CO 6	**0**	**3**	**3**	**0**	**2**	**0**	**0**	**0**	**0**	**1**
CO 7	**0**	**0**	**0**	**0**	**3**	**0**	**0**	**0**	**2**	**1**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

Assessment Types	**MODE OF ASSESSMENT**		
	A	**Continuous Comprehensive Assessment (CCA)**	
		Components	**Marks**
		Commitment, Punctuality & Professional Conduct	10
		Monthly Progress Reviews & Logbook Maintenance	15
		Skill Development & Application	15
		Interim Report	20
		Total	**60**
	B	**End Semester Evaluation (ESE)**	
		Components	**Marks**
		Feedback & Evaluation Report from Host Organization	40
		Skill Demonstration / Summary of Work Exposure	20
		Final Report / Learning Portfolio	25
		Domain Knowledge and Experience Communication (Presentation)	25
		Viva Voce	30
		Total	**140**

Note:

This assessment framework serves as a guiding structure for evaluating research internship performance. However, to remain responsive to the evolving needs of industry, academia, and society, the evaluation criteria may be revised periodically. Such revisions aim to enhance the relevance, effectiveness, and fairness of the overall assessment process.

	Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Technology and Applied Sciences/ Software Development and System Administration**
Programme	**B.Voc.(Honours) Software Development and System Administration**
Course Name	**On-the-Job Training**
Type of Course	**SDC**
Course Summary	On-the-Job Training (OJT) is designed to equip students with practical skills, workplace discipline, and industry exposure by actively engaging them in real- world professional environments. Conducted in collaboration with firms, industries, research institutions, or higher education establishments, OJT enables students to understand industry standards, apply academic knowledge, and perform job-specific tasks using contemporary tools and practices. The training must be undertaken in the student's own skill domain, aligned with the major area of study in their undergraduate program, to ensure relevance and coherence with their academic and career goals. The program also fosters essential workplace competencies such as communication, responsibility, adaptability, and teamwork. Furthermore, it offers students a platform for career exploration and networking, helping them evaluate potential career paths and align their aspirations with industry demands.

COURSE OUTCOMES (CO)

CO No:	Expected Course Outcome	Learning Domains	PO No:
	Upon the successful completion of the course, the student will be able to		
1	Demonstrate understanding of industry operations, standards, and professional expectations through direct exposure to workplace environments.	Ap	1,3,6,10
2	Apply job-specific skills effectively in real-world tasks and responsibilities within the assigned industry setting.	S	2,4,5,10

3	Integrate academic knowledge with practical applications to solve work-related challenges and contribute to organizational goals.	An	1,2,3,6
4	Exhibit essential workplace competencies such as punctuality, accountability, communication, teamwork, and adaptability.	S	4,5,8,9
5	Identify and evaluate potential career opportunities by reflecting on their internship experiences and professional interactions.	E	1,9,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

Assessment Types	**MODE OF ASSESSMENT**		
	A. Continuous Comprehensive Assessment (CCA)		
		Components	**Marks**
		Feedback from the hosting organization	5
		Internal Supervisor feedback	10
		Total	**15**
	B. End Semester Evaluation (ESE)		
		Components	**Marks**
		Presentation	10
		Report	10
		Viva Voce	15
		Total	**35**